

# Copyright Notice

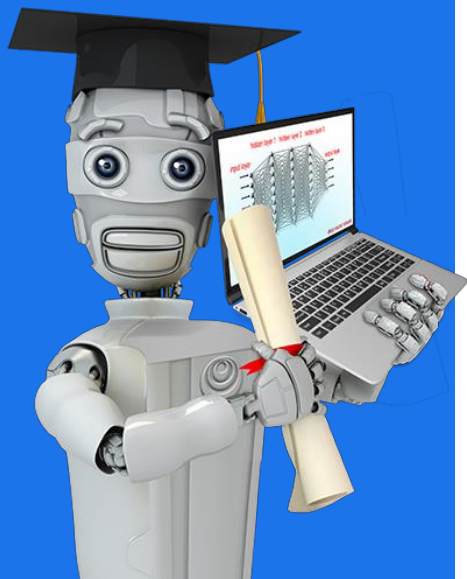
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Stanford  
ONLINE

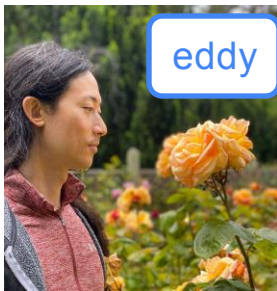
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# Machine Learning

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Welcome!



eddy



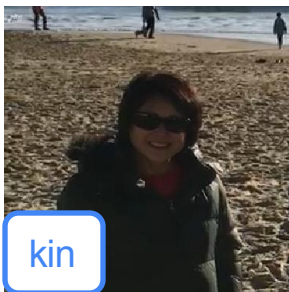
aarti



geoff



Ivy



kin



robert



andres



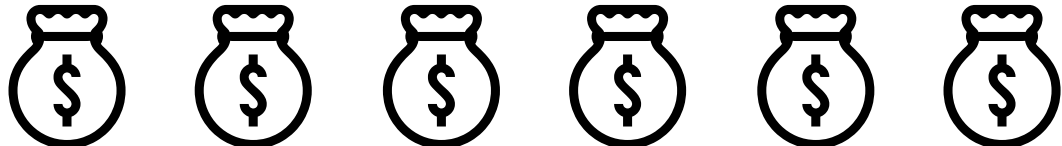
daniel

Re: Urgent Information :)

External

Spam x

Congratulations!  
You've won  
a million dollars!



Compose

Mail

Inbox

Starred

Snoozed

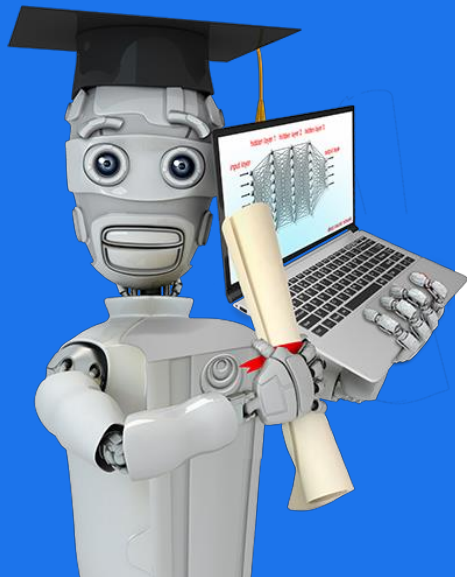
Sent

Drafts



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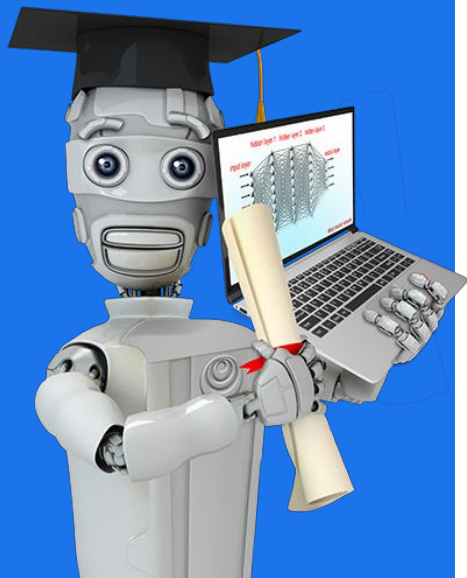
# Machine Learning

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## Applications of Machine Learning

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# Machine Learning Overview

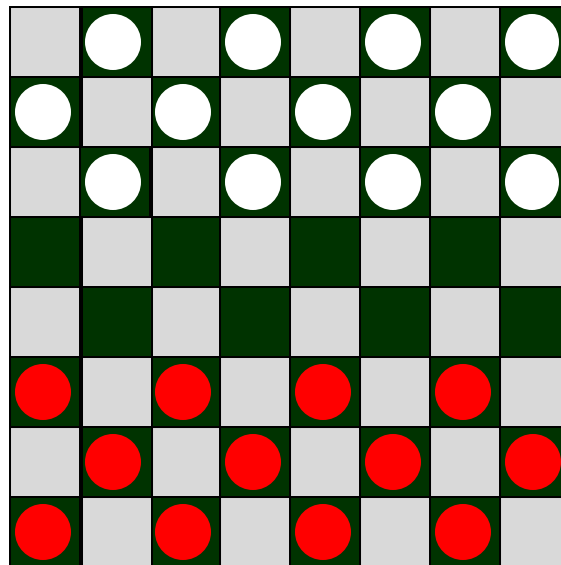
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What is  
Machine Learning?

# Machine learning

“Field of study that gives computers the ability to learn without being explicitly programmed.”

Arthur Samuel (1959)



# Question

If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

☐ Would have made it better

→ ☒ Would have made it worse

# Machine learning algorithms

*rapid advancements*

*used most in real-world applications*

- Supervised learning ← *course 1, 2*

- Unsupervised learning ←

*course 3*

- Recommender systems

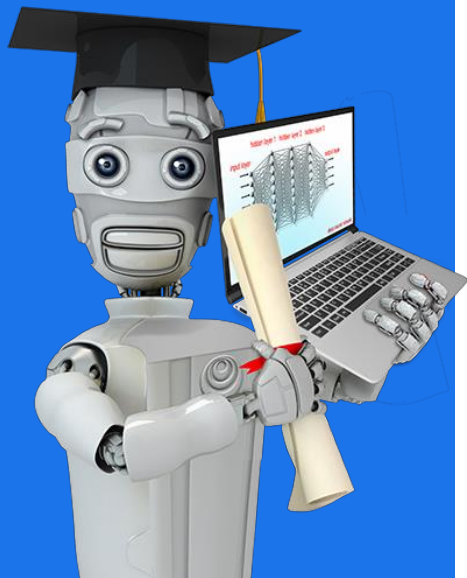
- Reinforcement learning

Practical advice for applying learning algorithms



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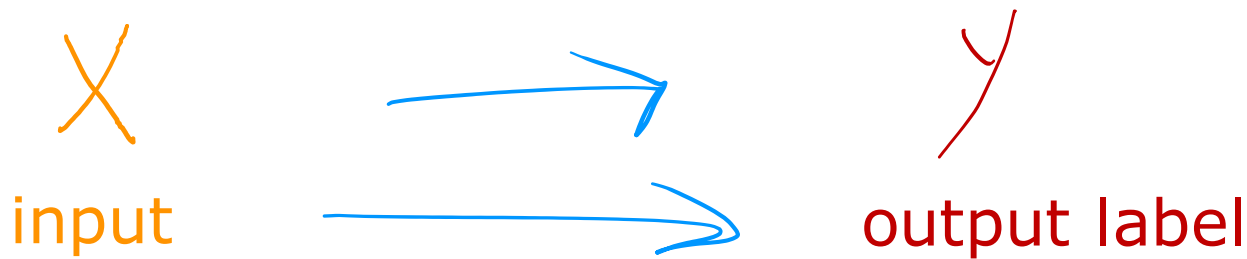


# Machine Learning Overview

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## Supervised Learning Part 1

# Supervised learning

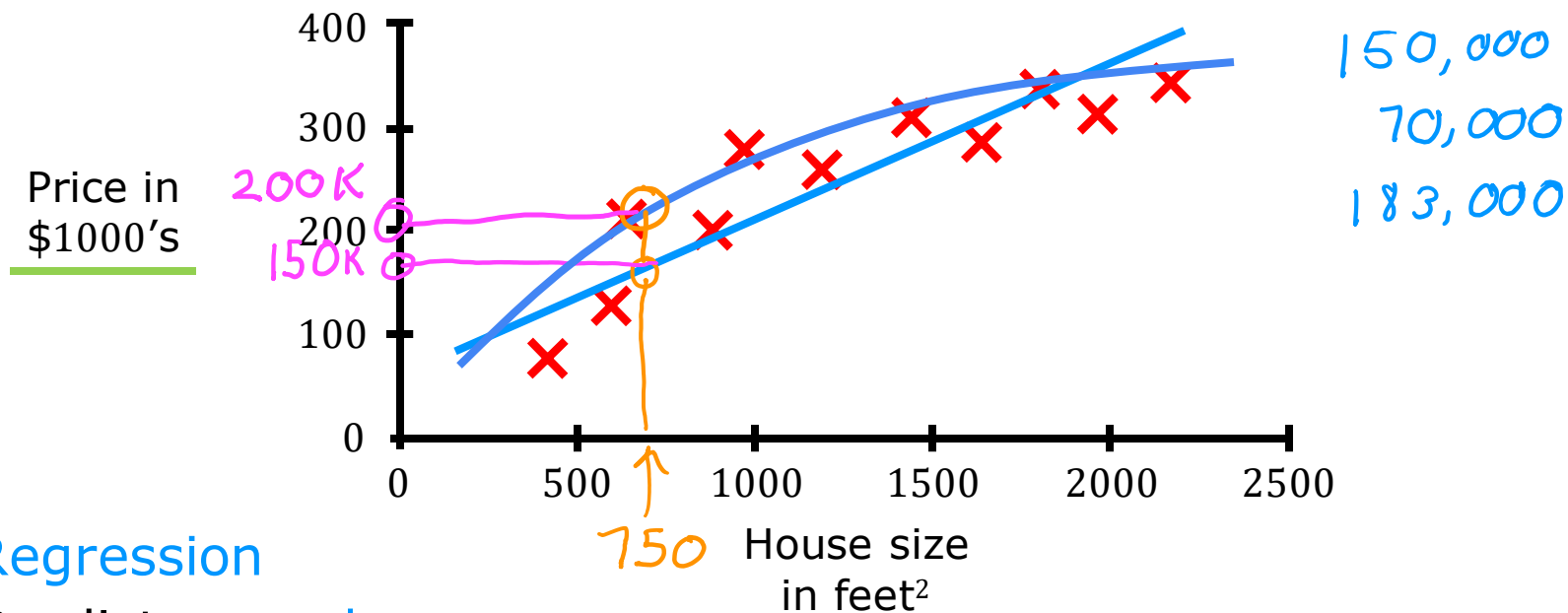


Learns from being given “right answers”

| Input (X)         |   | Output (Y)             | Application         |
|-------------------|---|------------------------|---------------------|
| email             | → | spam? (0/1)            | spam filtering      |
| audio             | → | text transcripts       | speech recognition  |
| English           | → | Spanish                | machine translation |
| ad, user info     | → | click? (0/1)           | online advertising  |
| image, radar info | → | position of other cars | self-driving car    |
| image of phone    | → | defect? (0/1)          | visual inspection   |



# Regression: Housing price prediction



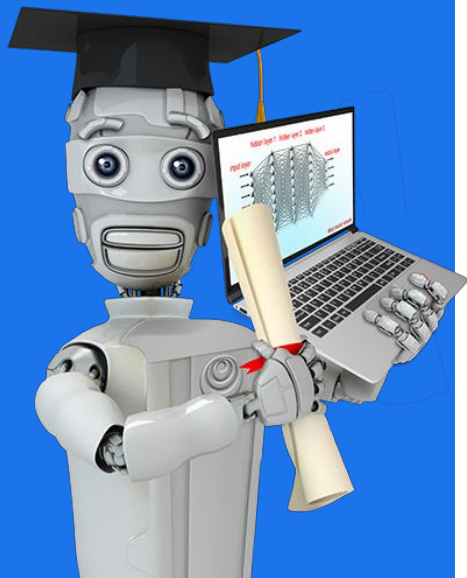
Regression

Predict a **number**

**infinitely** many possible outputs

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# Machine Learning Overview

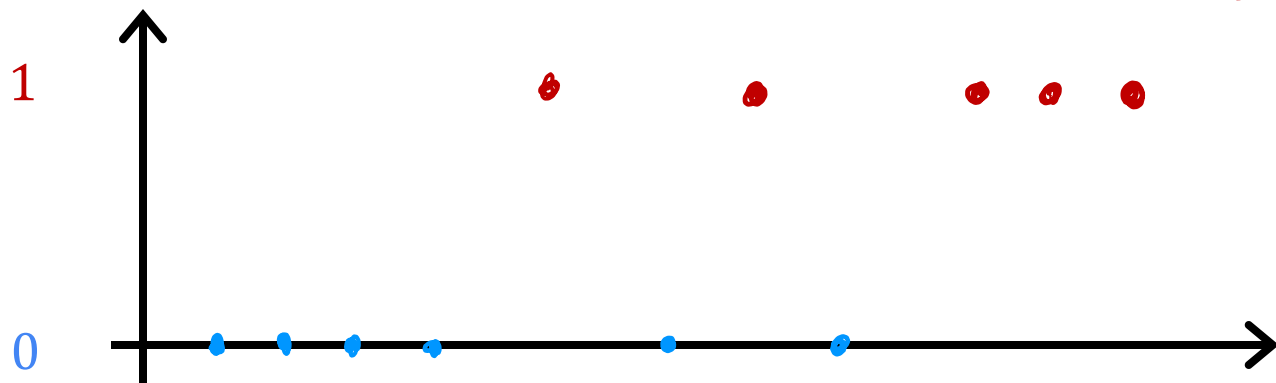
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## Supervised Learning Part 2

# Classification: Breast cancer detection

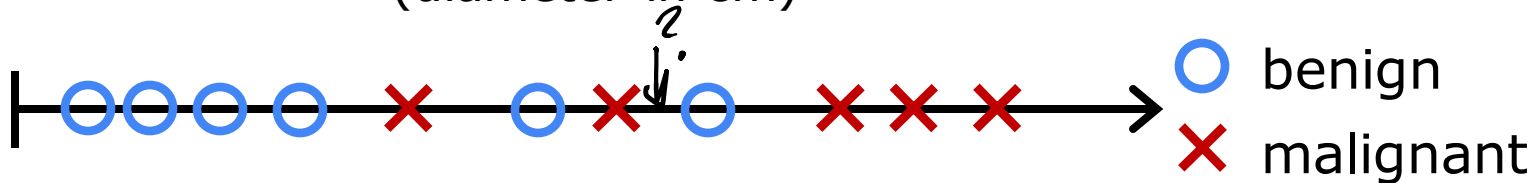


malignant benign



| size | diagnosis |
|------|-----------|
| 2    | ○         |
| 5    | ×         |
| 1    | ○         |
| 7    | ×         |
| ⋮    |           |

tumor size  $x$   
(diameter in cm)



# Classification: Breast cancer detection

- benign
- ✗ malignant *type 1*
- △ malignant type 2



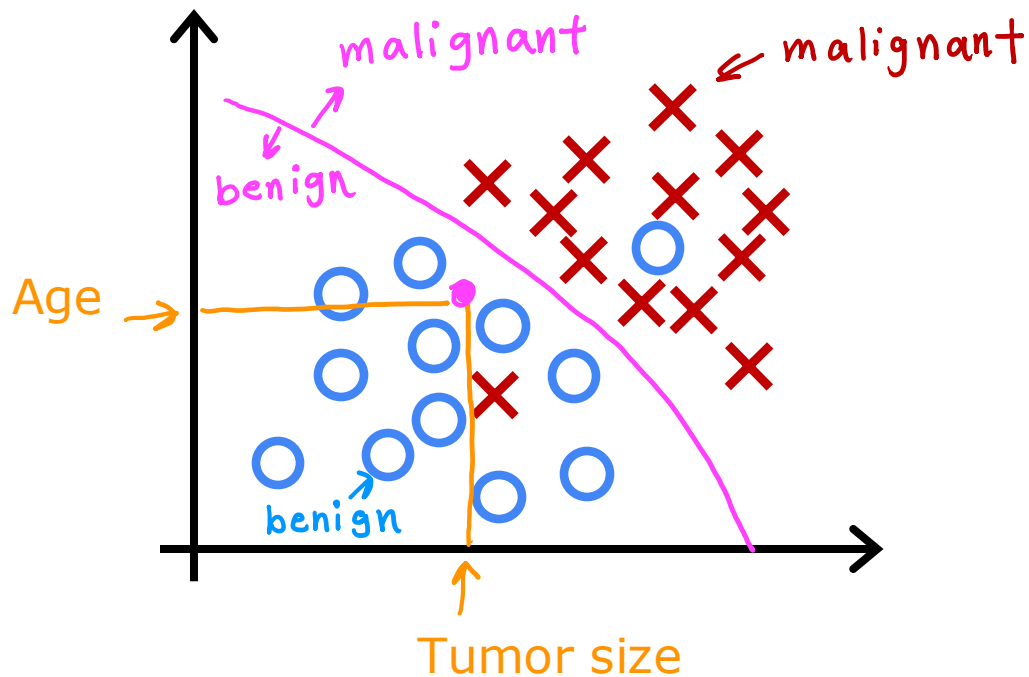
*class category*

Classification

predict *categories*    *cat dog benign malignant 0, 1, 2*

*small number* of possible outputs

# Two or more inputs



# Supervised learning

Learns from being given “right answers”

## Regression

Predict a number

infinitely many possible outputs

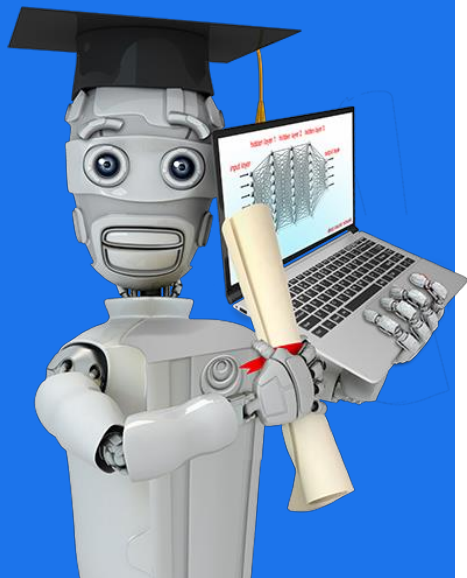
## Classification

predict categories

small number of possible outputs

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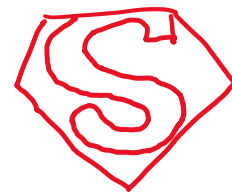
# Machine Learning Overview

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## Unsupervised Learning Part 1

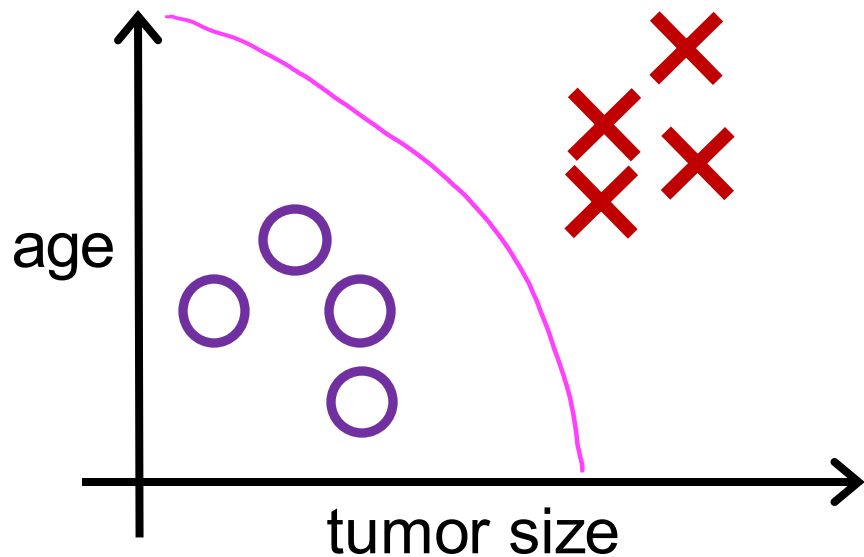
Previous: Supervised learning

Now: Unsupervised learning

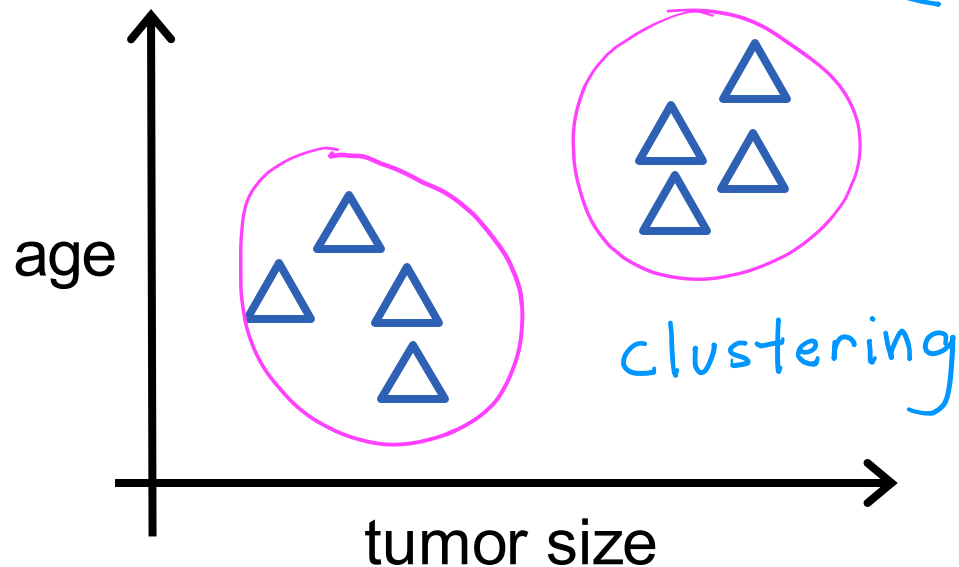




Supervised learning  
Learn from data **labeled**  
with the “**right answers**”



Unsupervised learning  
Find something interesting  
in **unlabeled** data.

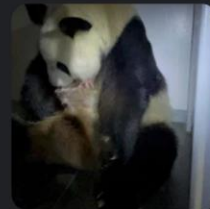


# Clustering: Google news



Giant **panda** gives birth to rare **twin** cubs at Japan's oldest **zoo**

USA TODAY · 6 hours ago



- Giant **panda** gives birth to **twin** cubs at Japan's oldest **zoo**

CBS News · 7 hours ago

- Giant **panda** gives birth to **twin** cubs at Tokyo's Ueno **Zoo**

WHBL News · 16 hours ago

- A Joyful Surprise at Japan's Oldest **Zoo**: The Birth of **Twin Pandas**

The New York Times · 1 hour ago

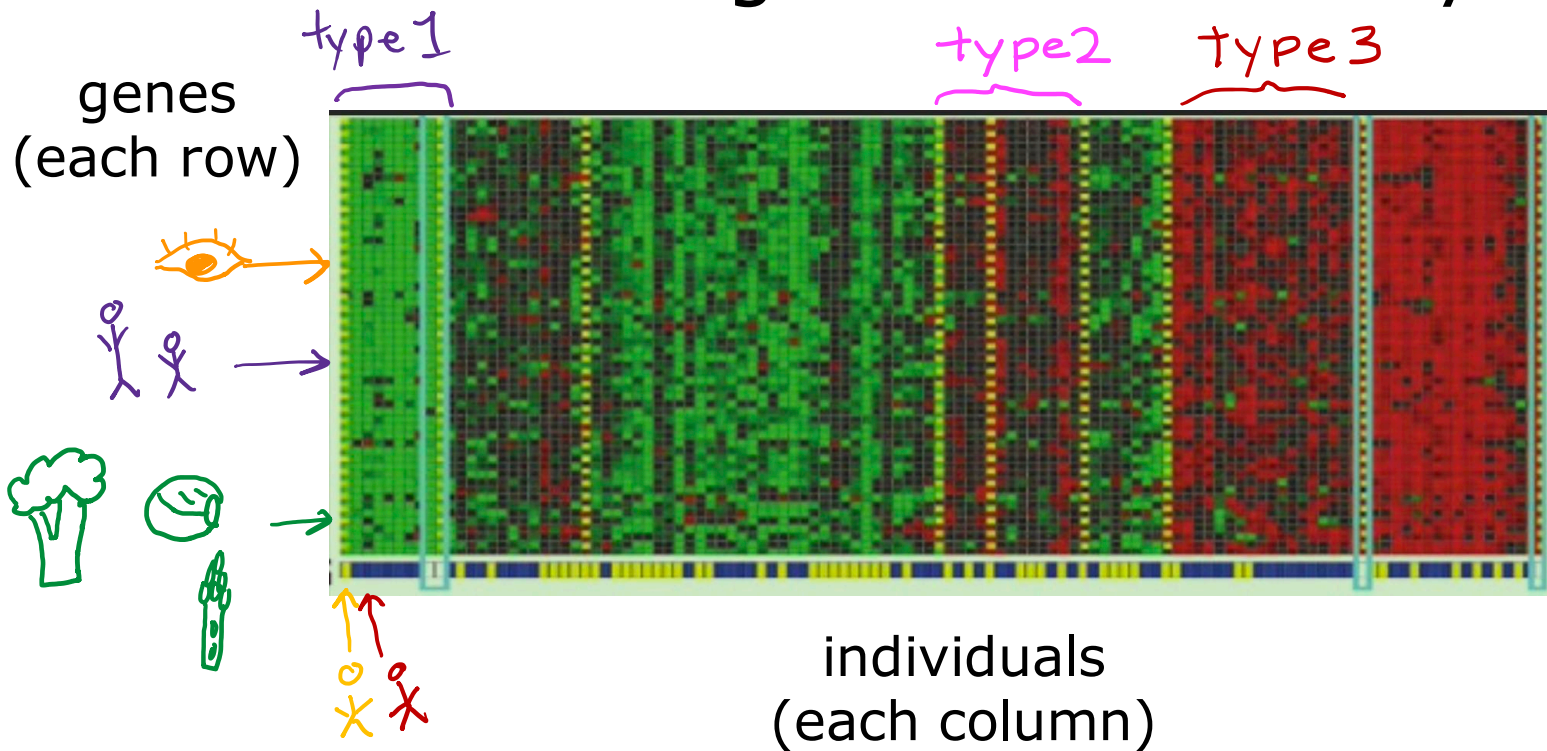
- **Twin Panda** Cubs Born at Tokyo's Ueno **Zoo**

PEOPLE · 6 hours ago

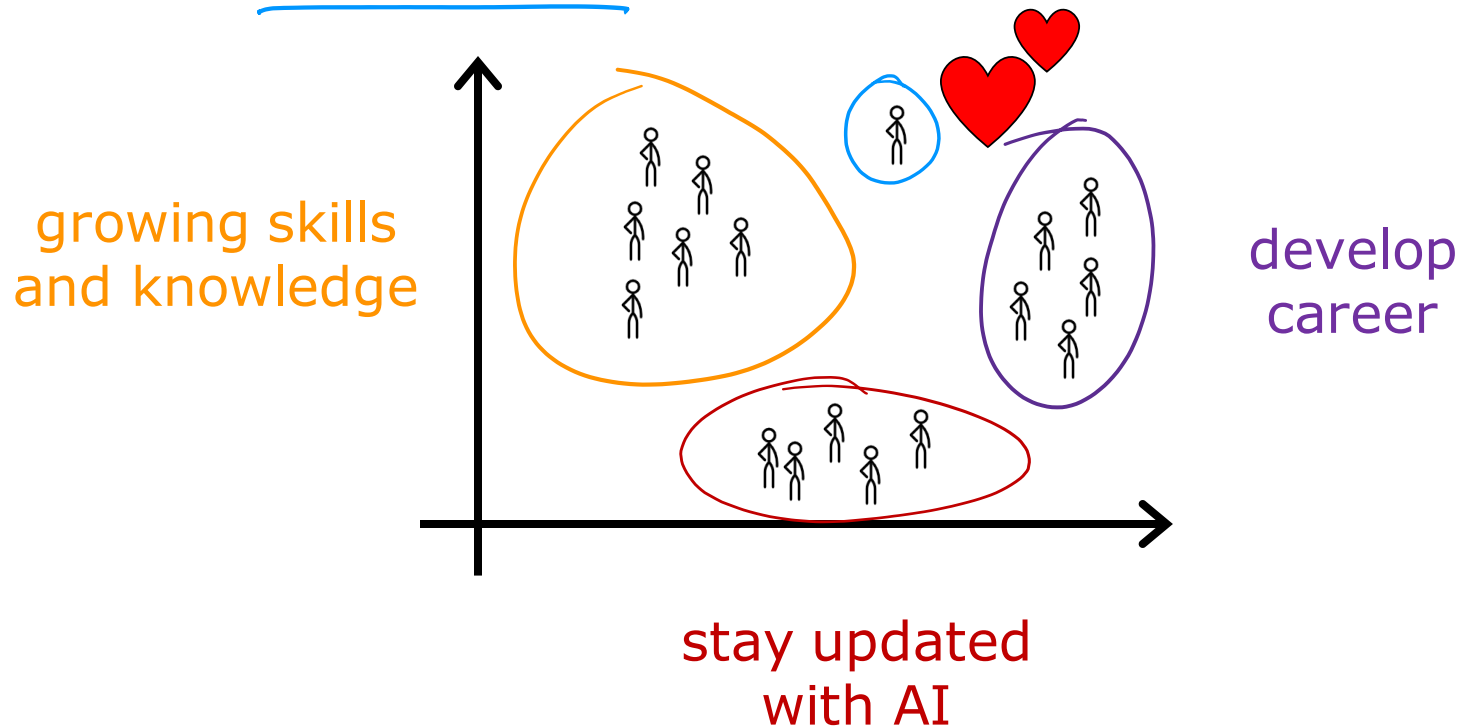
 [View Full Coverage](#)



# Clustering: DNA microarray

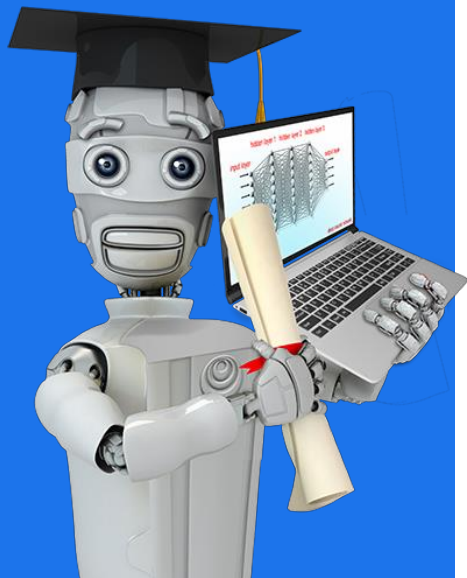


# Clustering: Grouping customers



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# Machine Learning Overview

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## Unsupervised Learning Part 2

# Unsupervised learning

Data only comes with inputs  $x$ , but not output labels  $y$ .  
Algorithm has to find **structure** in the data.

## Clustering

Group similar data points together.

## Dimensionality reduction

Compress data using fewer numbers.

## Anomaly detection

Find unusual data points.

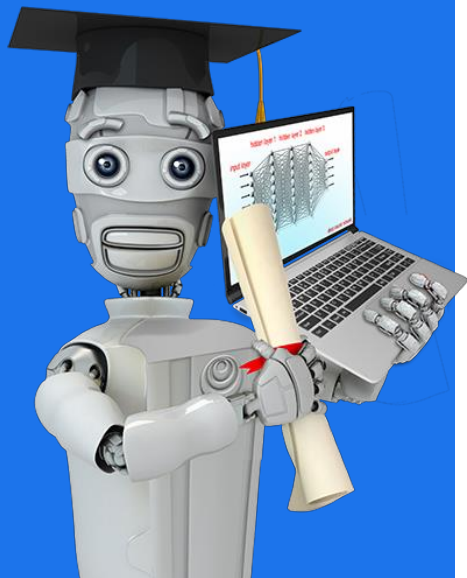
# Question

Of the following examples, which would you address using an **unsupervised** learning algorithm?

-  ☐ Given email labeled as spam/not spam, learn a spam filter.
-  ☒ Given a set of news articles found on the web, group them into sets of articles about the same story.
-  ☒ Given a database of customer data, automatically discover market segments and group customers into different market segments.
-  ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

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# Machine Learning Overview

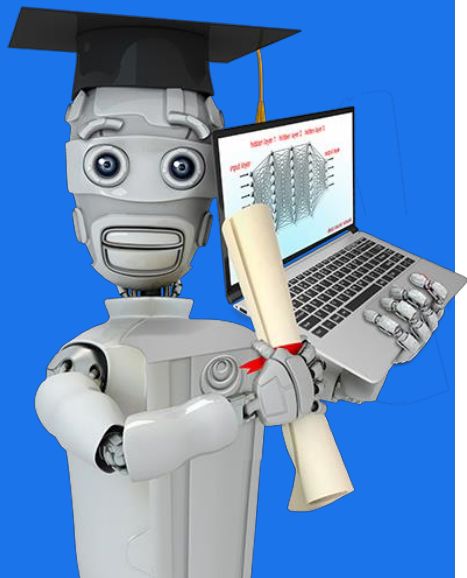
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## Jupyter Notebooks



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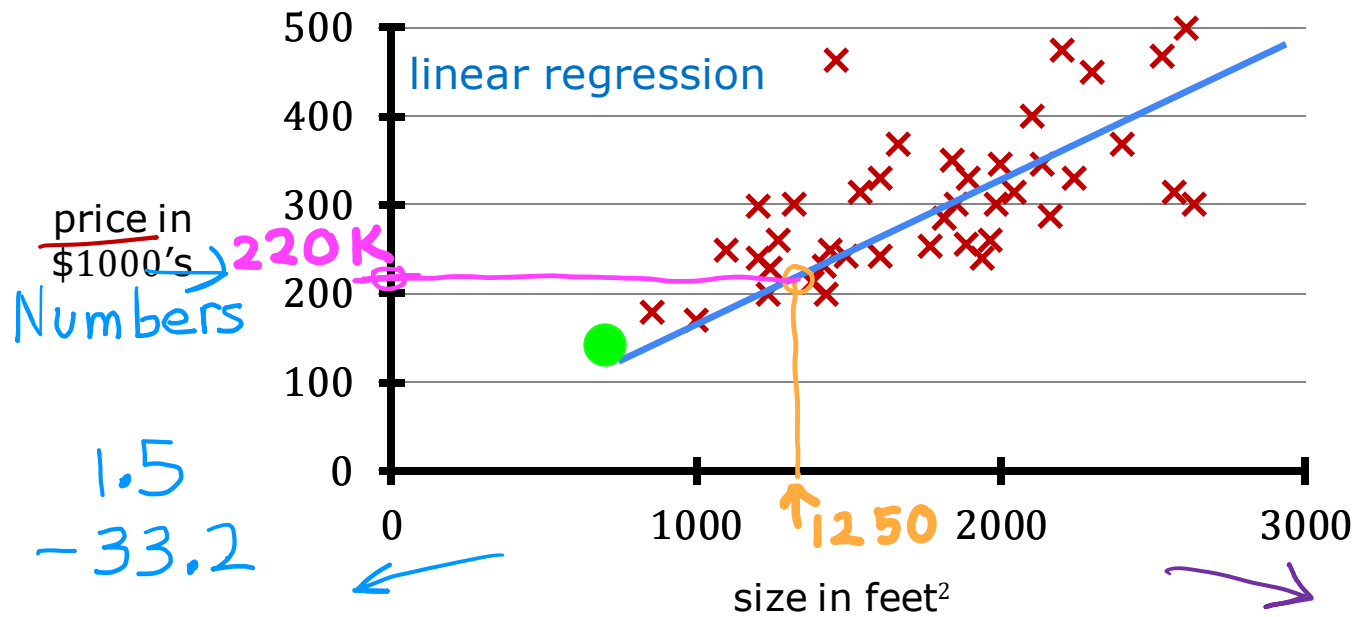


# Linear Regression with One Variable

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## Linear Regression Model Part 1

# House sizes and prices



categories

cat } 2  
dog }

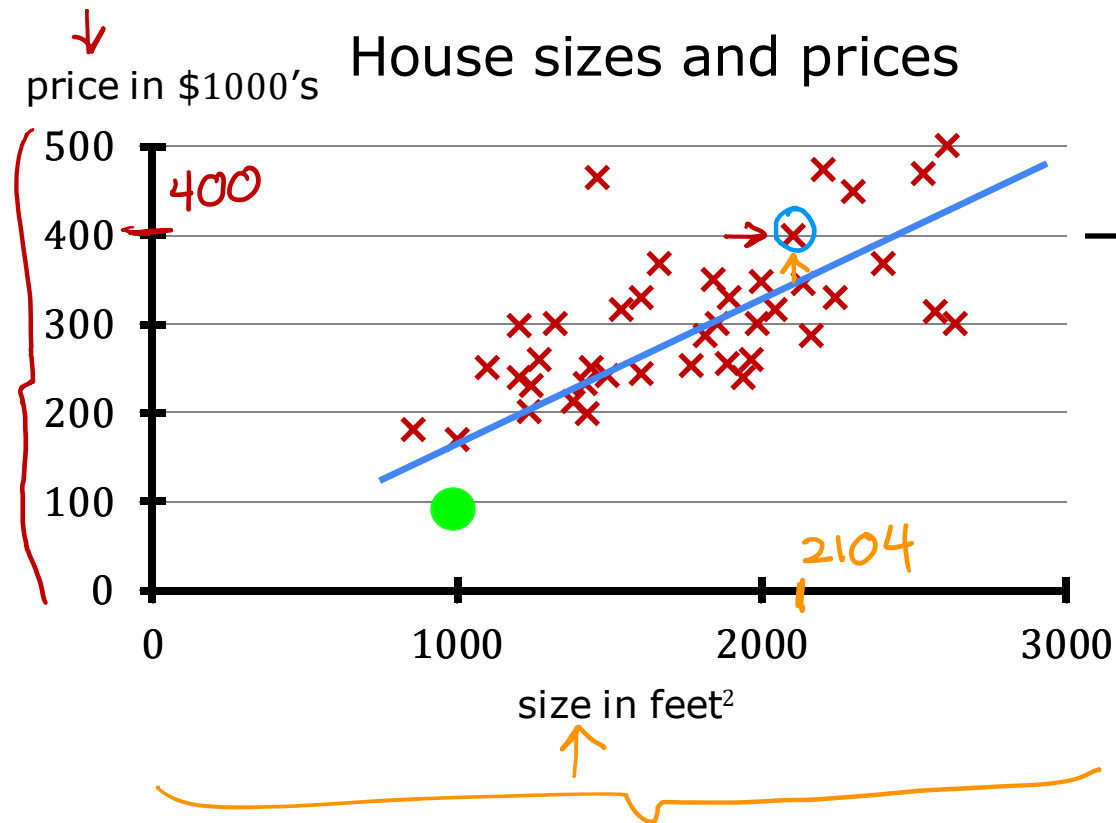
disease  10

Regression model  
Predicts numbers  
Infinitely many possible outputs

Supervised learning model  
Data has "right answers"

Classification model  
Predicts categories  
Small number of possible outputs

# House sizes and prices



Data table

| size in feet <sup>2</sup> | price in \$1000's |
|---------------------------|-------------------|
| 2104                      | 400               |
| 1416                      | 232               |
| 1534                      | 315               |
| 852                       | 178               |
| ...                       | ...               |
| 3210                      | 870               |

# Terminology

Training Data used to train the model

set:  $x$   $y$   
→ size in feet<sup>2</sup> → price in \$1000's

|      |      |     |
|------|------|-----|
| (1)  | 2104 | 400 |
| (2)  | 1416 | 232 |
| (3)  | 1534 | 315 |
| (4)  | 852  | 178 |
| ...  | ...  | ... |
| (47) | 3210 | 870 |

$m = 47$

$$x^{(1)} = 2104 \quad y^{(1)} = 400$$

$$(x^{(1)}, y^{(1)}) = (2104, 400)$$

$$x^{(2)} = 1416 \quad x^{(2)} \neq x^2 \text{ not exponent}$$

Notation:

$x$  = "input" variable  
feature

$y$  = "output" variable  
"target" variable

$m$  = number of training examples

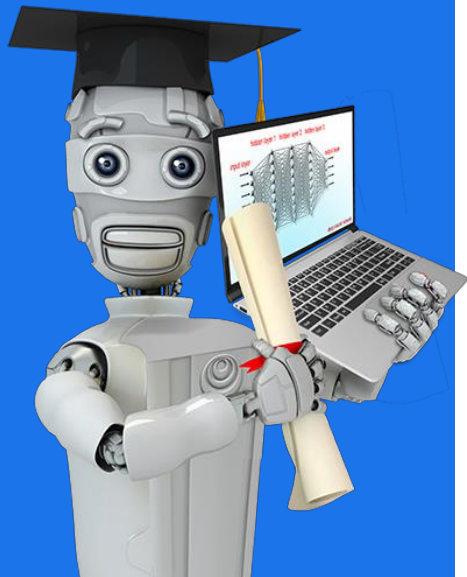
$(x, y)$  = single training example

$$(x^{(i)}, y^{(i)})$$

$(x^{(i)}, y^{(i)})$  =  $i^{\text{th}}$  training example  
index (1<sup>st</sup>, 2<sup>nd</sup>, 3<sup>rd</sup> ...)

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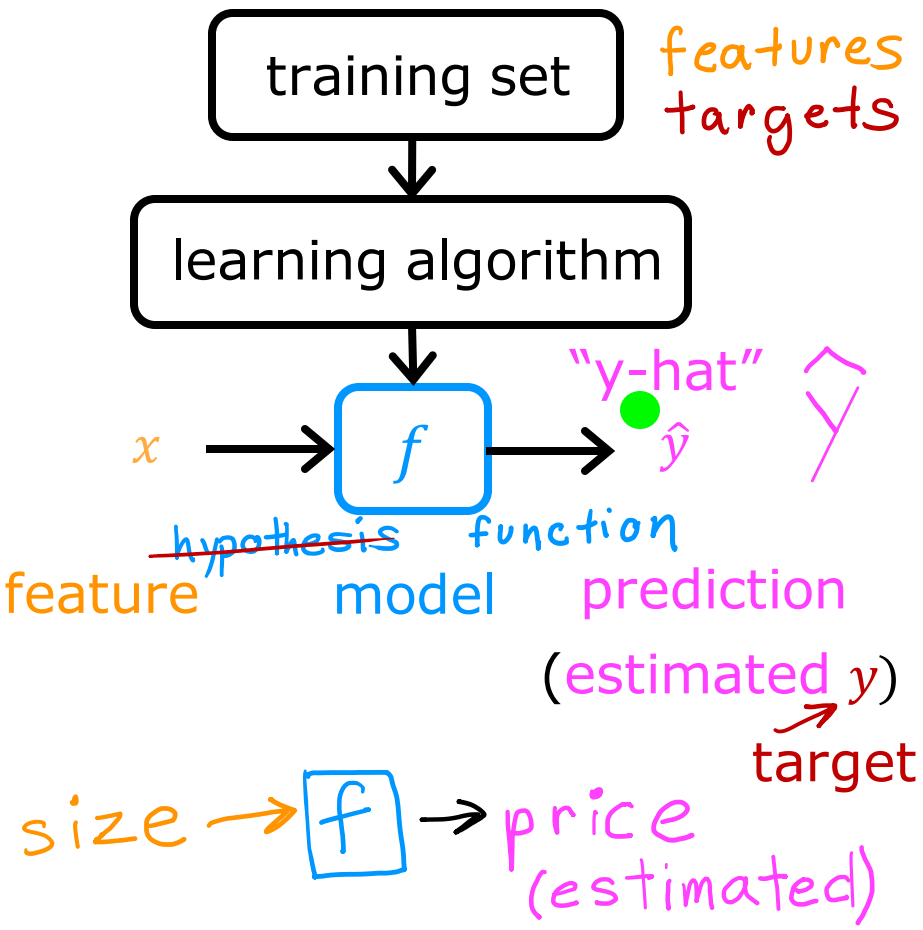
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# Linear Regression with One Variable

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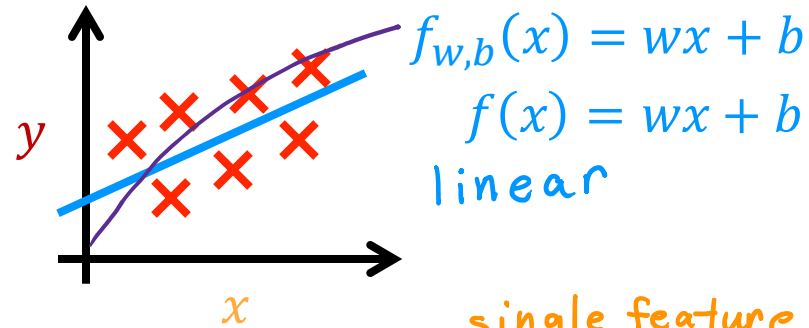
## Linear Regression Model Part 2



How to represent  $f$ ?

$$f_{w,b}(x) = wx + b$$

$f(x)$

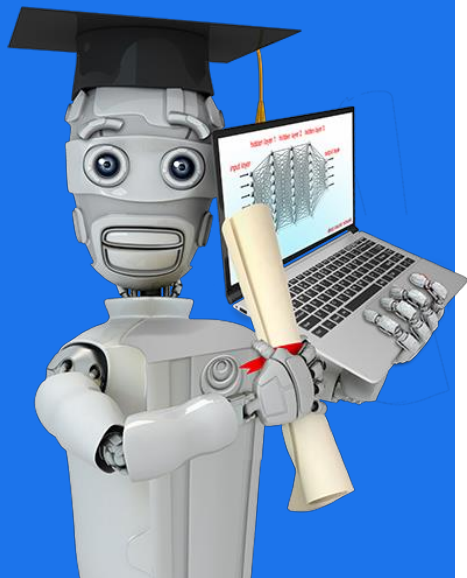


Linear regression with *one* variable.  
*size*

Univariate linear regression.  
*one variable*

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# Linear Regression with One Variable

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## Cost Function

# Training set

| <sup>features</sup><br>size in feet <sup>2</sup> ( $x$ ) | <sup>targets</sup><br>price \$1000's ( $y$ ) |
|----------------------------------------------------------|----------------------------------------------|
| 2104                                                     | 460                                          |
| 1416                                                     | 232                                          |
| 1534                                                     | 315                                          |
| 852                                                      | 178                                          |
| ...                                                      | ...                                          |

Model:  $f_{w,b}(x) = wx + b$

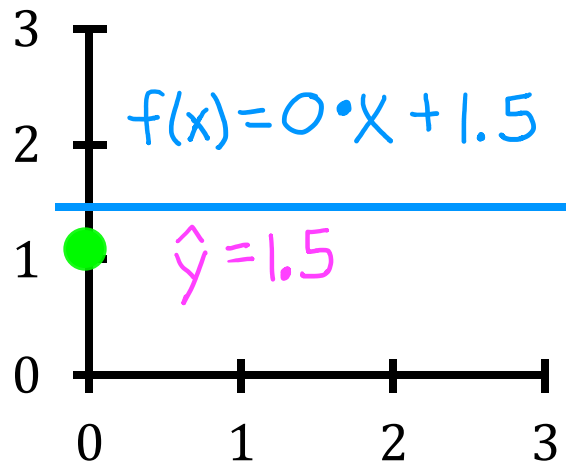
$w, b$ : parameters  
coefficients  
weights

What do  $w, b$  do?

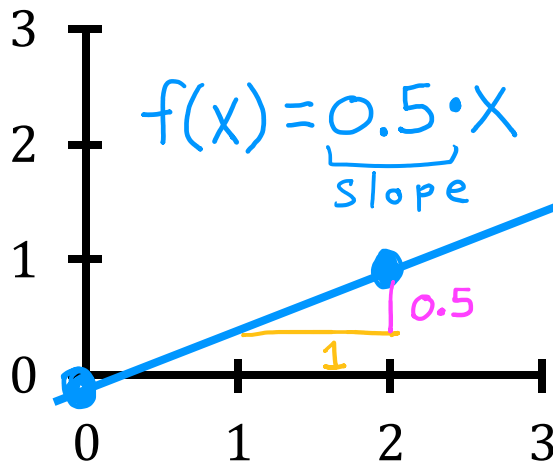


$$\underline{f_{w,b}}(x) = wx + b$$

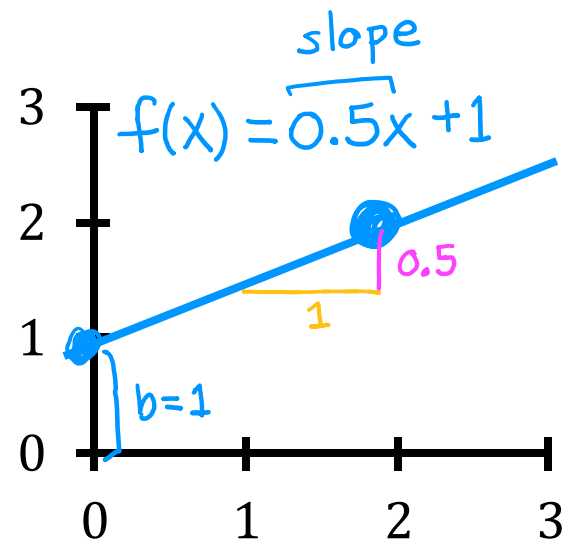
$f(x)$



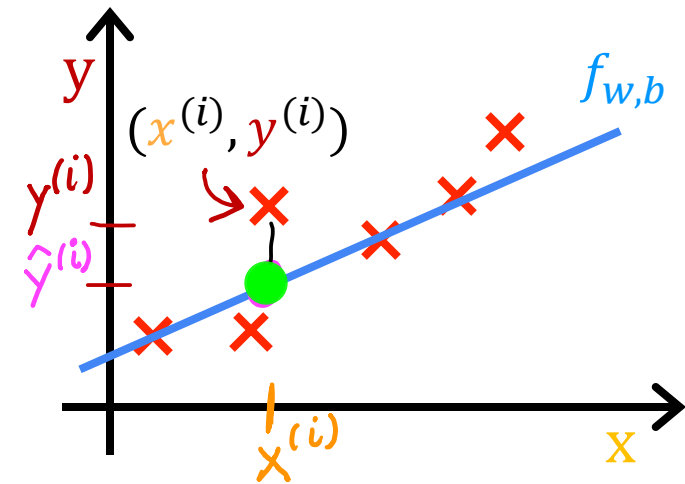
→  $w = 0$   
 →  $b = 1.5$   
 (y-intercept)



→  $w = 0.5$   
 →  $b = 0$



→  $w = 0.5$   
 →  $b = 1$



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)}) \quad \leftarrow$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function: Squared error cost function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m \left( \hat{y}^{(i)} - y^{(i)} \right)^2$$

error

$m$  = number of training examples

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m \left( f_{w,b}(x^{(i)}) - y^{(i)} \right)^2$$

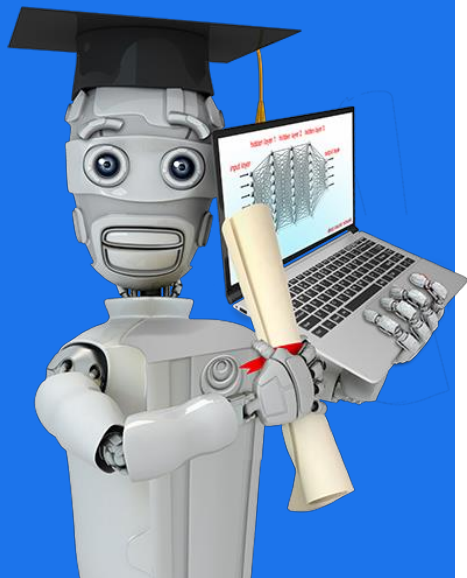
intuition (next!)

Find  $w, b$ :

$\hat{y}^{(i)}$  is close to  $y^{(i)}$  for all  $(x^{(i)}, y^{(i)})$ .

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# Linear Regression with One Variable

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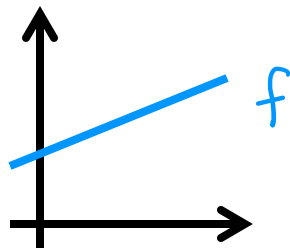
## Cost Function Intuition

model:

$$\underline{f_{w,b}(x) = wx + b}$$

parameters:

$$\underline{w, b}$$



cost function:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

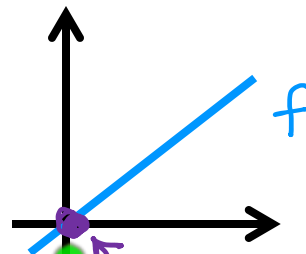
$$\underset{w, b}{\text{minimize}} J(w, b)$$

simplified

$$f_w(x) = \underline{wx}$$

$$b = \emptyset$$

w



$$\underline{J(w)} = \frac{1}{2m} \sum_{i=1}^m (\underline{f_w(x^{(i)})} - y^{(i)})^2$$

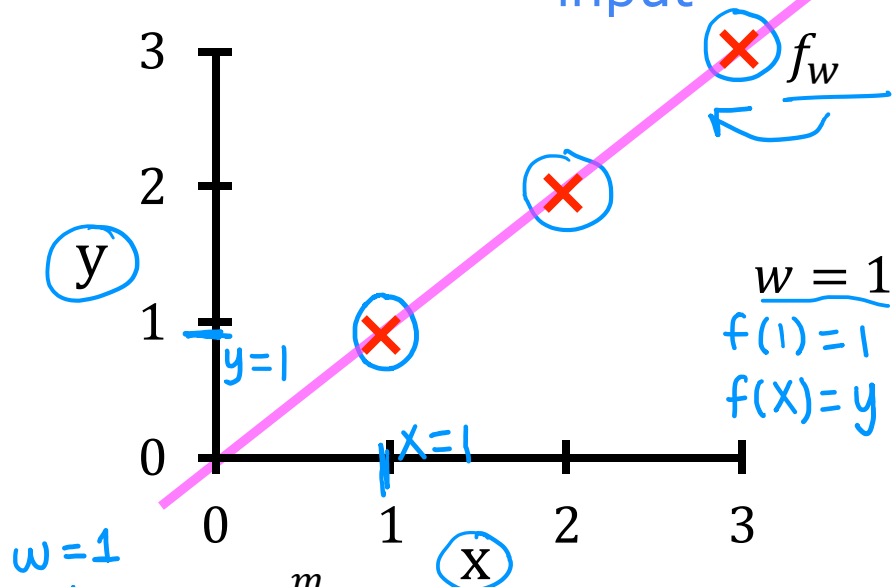
$$\underset{\underline{w}}{\text{minimize}} \underline{J(w)}$$

$$wx^{(i)}$$

→  $f_w(x)$

(for fixed  $w$ , function of  $x$ )

input

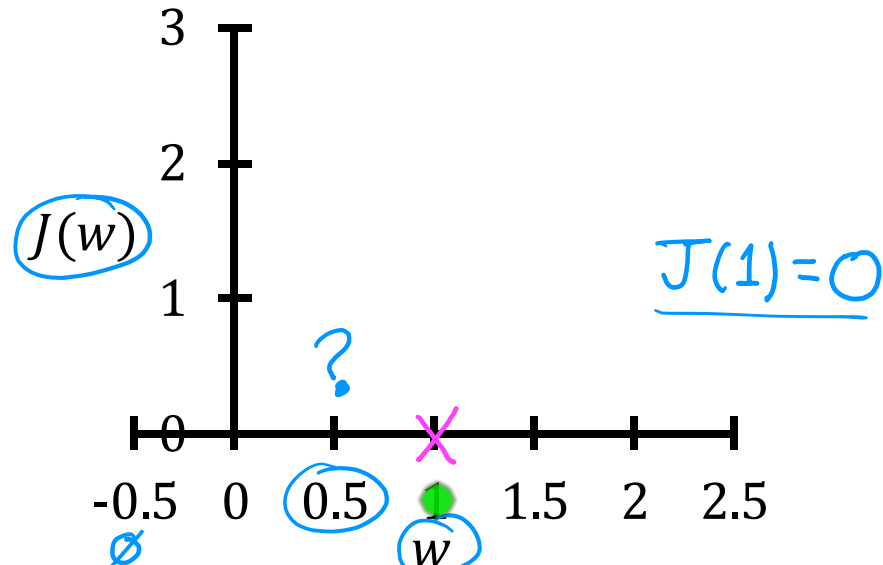


$$J(w) = \frac{1}{2m} \sum_{i=1}^m (f_w(x^{(i)}) - y^{(i)})^2 = \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} - y^{(i)})^2 = \frac{1}{2m} (0^2 + 0^2 + 0^2) = 0$$

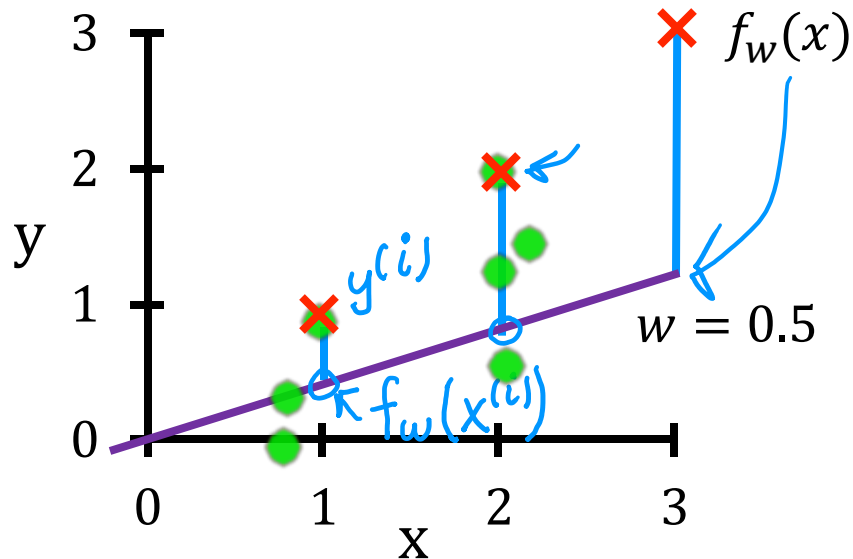
$J(w)$

(function of  $w$ )

parameter

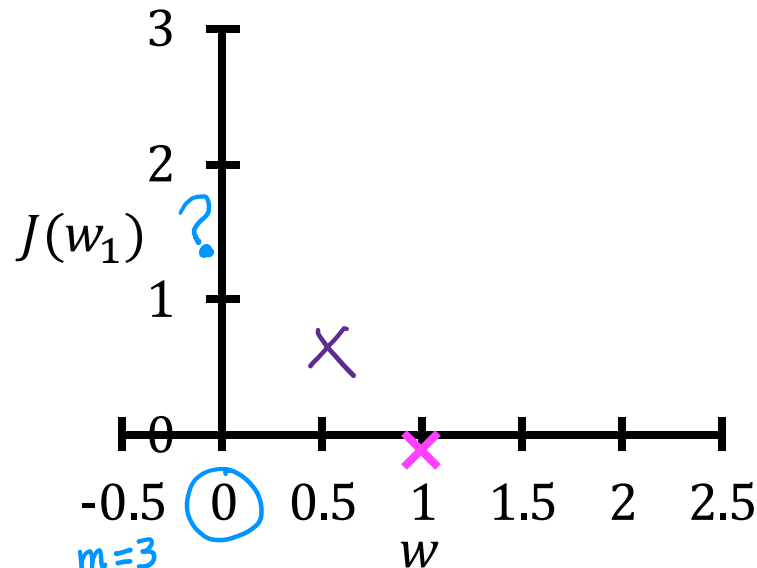


$f_w(x)$   
(function of  $x$ )

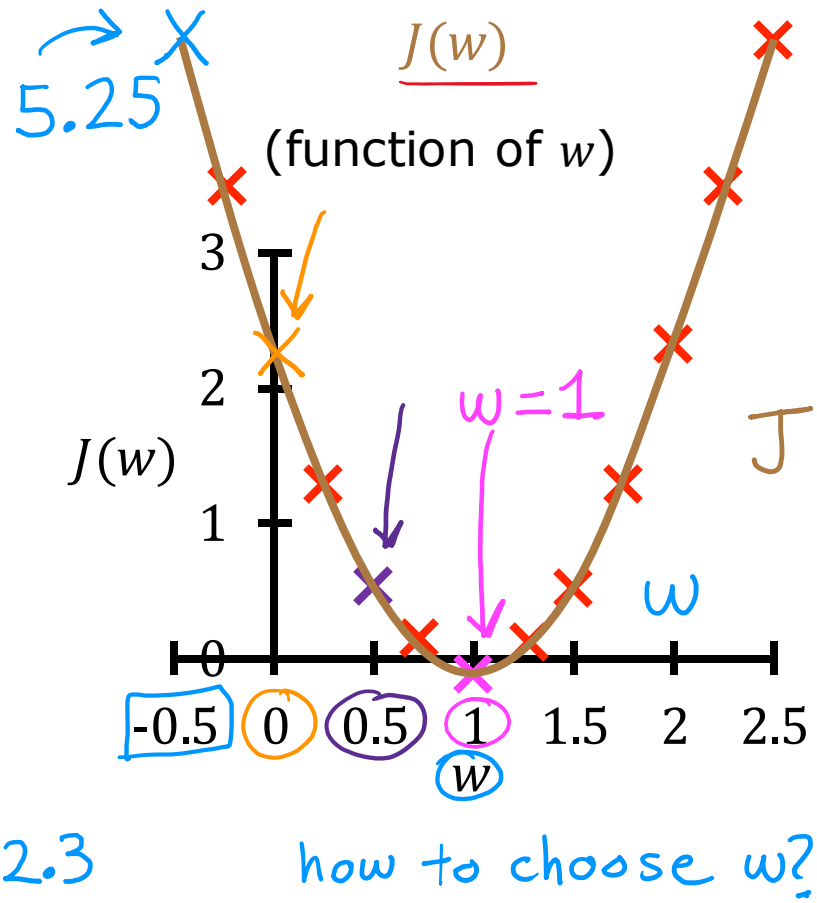
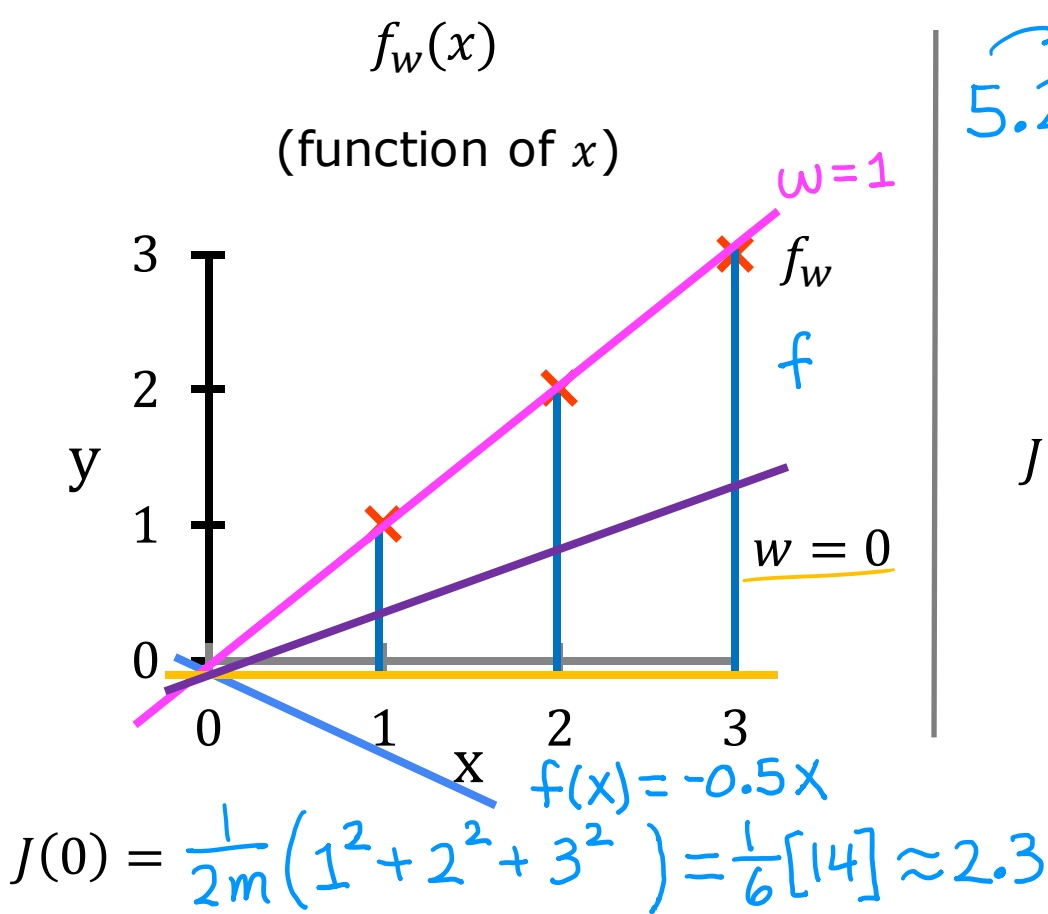


$J(w)$

(function of  $w$ )



$$J(0.5) = \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] = \frac{1}{2 \times 3} [3.5] = \frac{3.5}{6} \approx 0.58$$

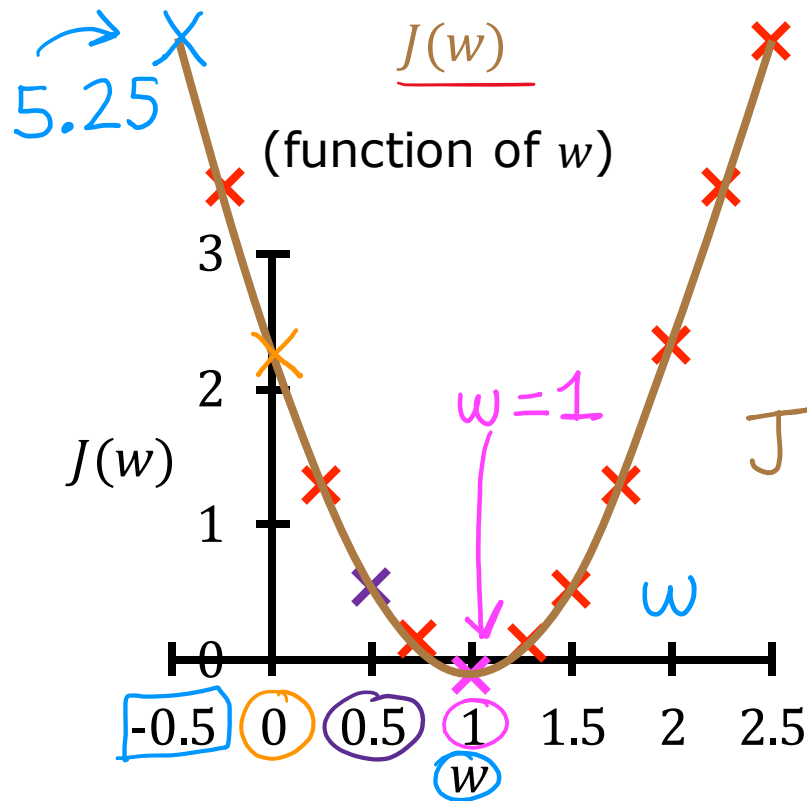


goal of linear regression:

$$\underset{w}{\text{minimize}} J(w)$$

general case:

$$\underset{w,b}{\text{minimize}} J(w, b)$$

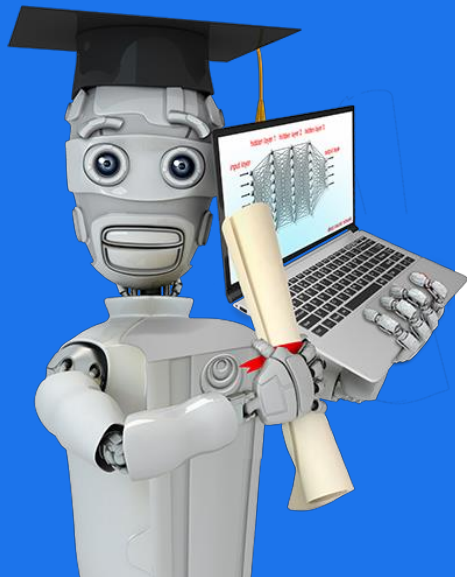


choose  $w$  to minimize  $J(w)$



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# Linear Regression with One Variable

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Visualizing  
the Cost Function

Model  $f_{w,b}(x) = wx + b$

Parameters

$w, b$

~~before:  $b=0$~~

Cost Function

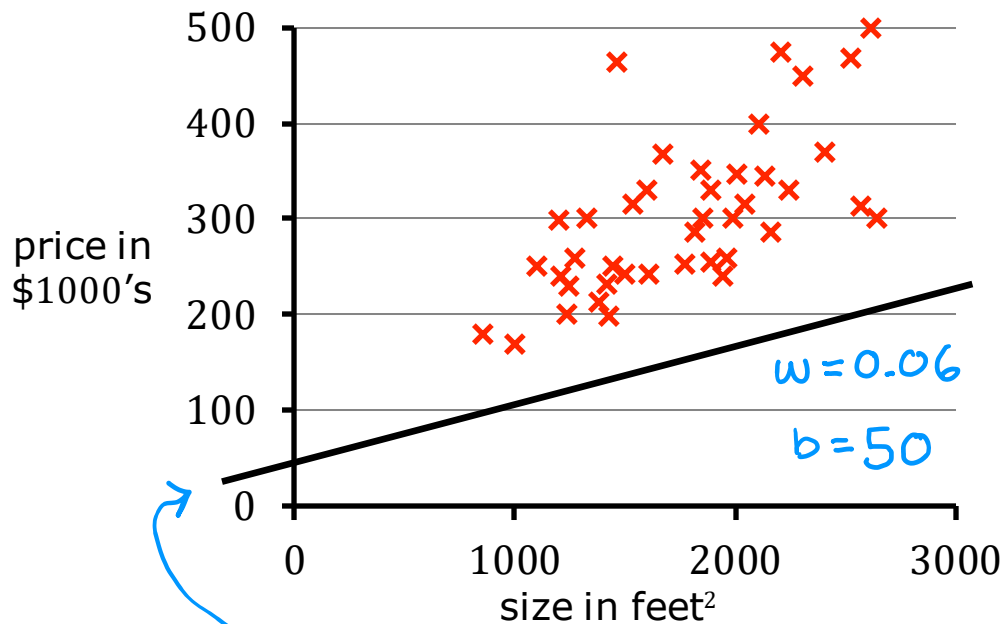
$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Objective

minimize  $J(w, b)$   
 $w, b$

$f_{w,b}$

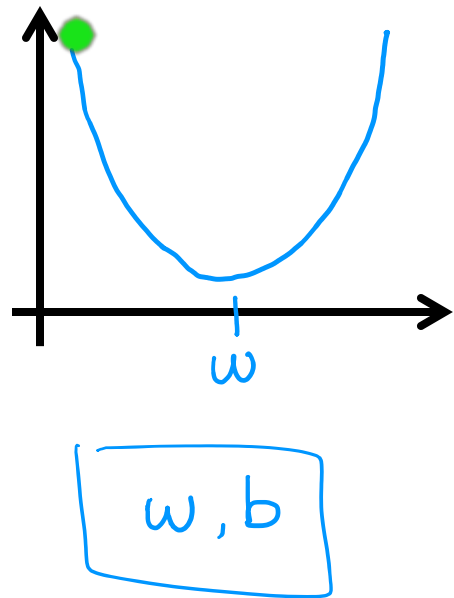
(function of  $x$ )

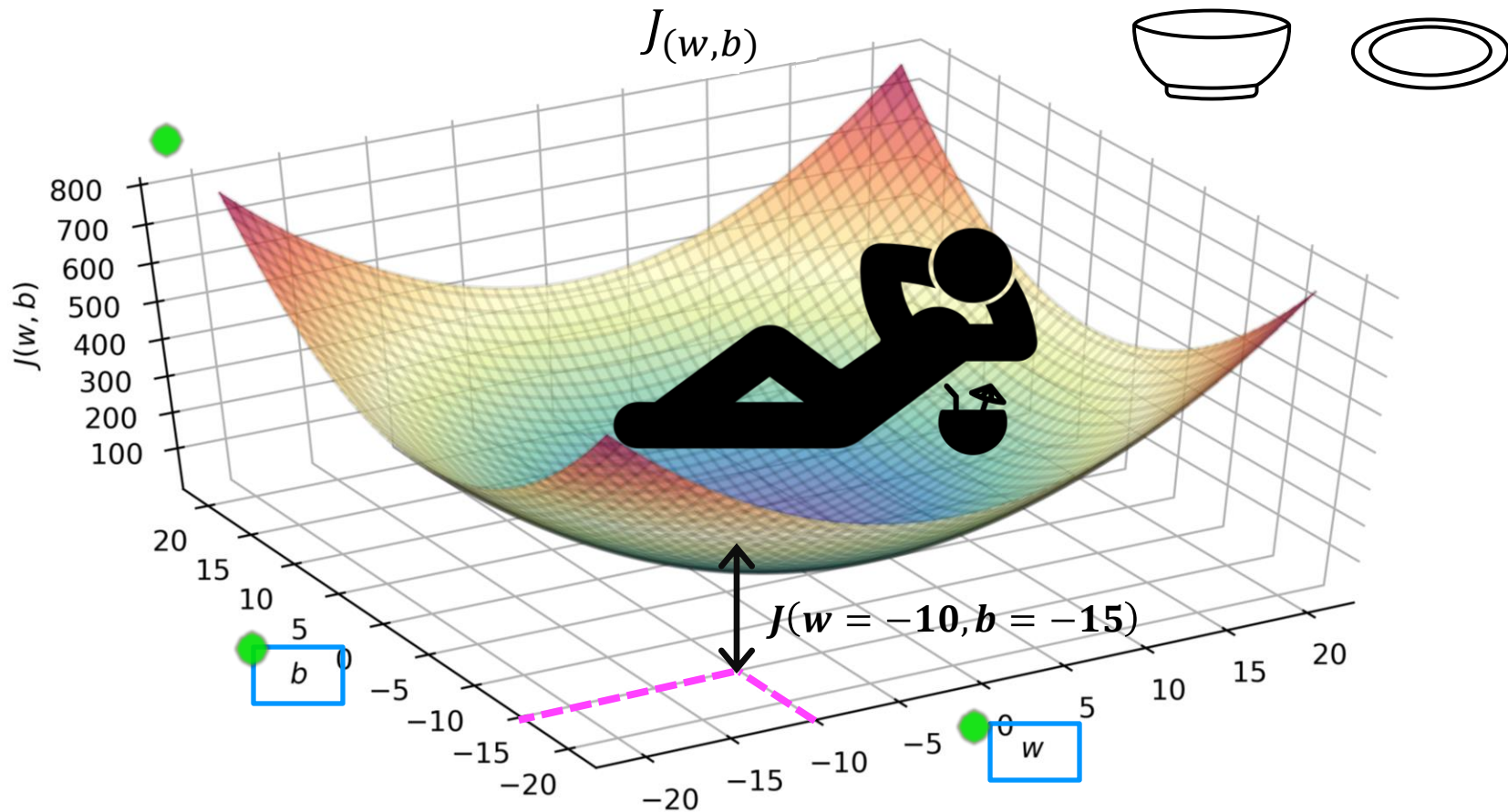


$$f_{w,b}(x) = 0.06x + 50$$

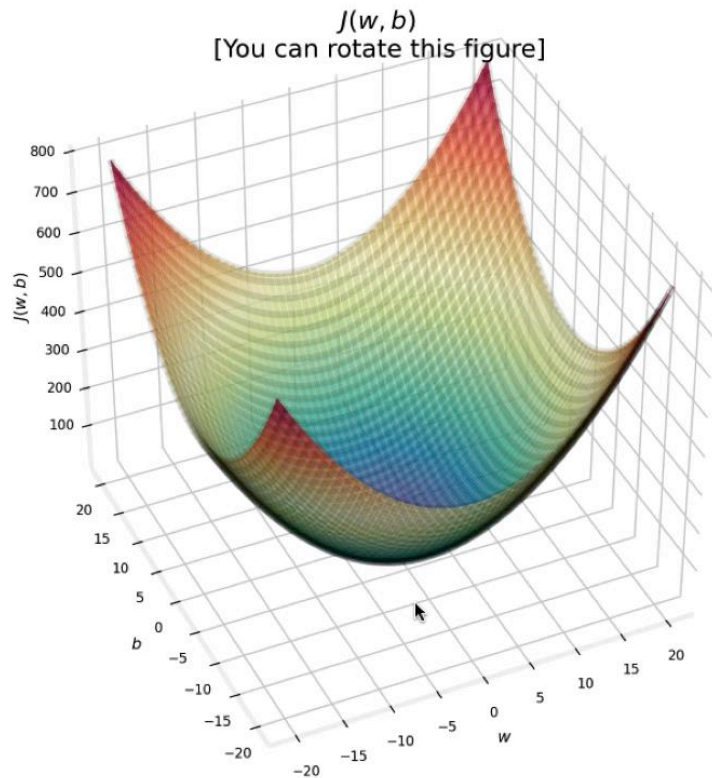
$J$

(function of  $w, b$ )



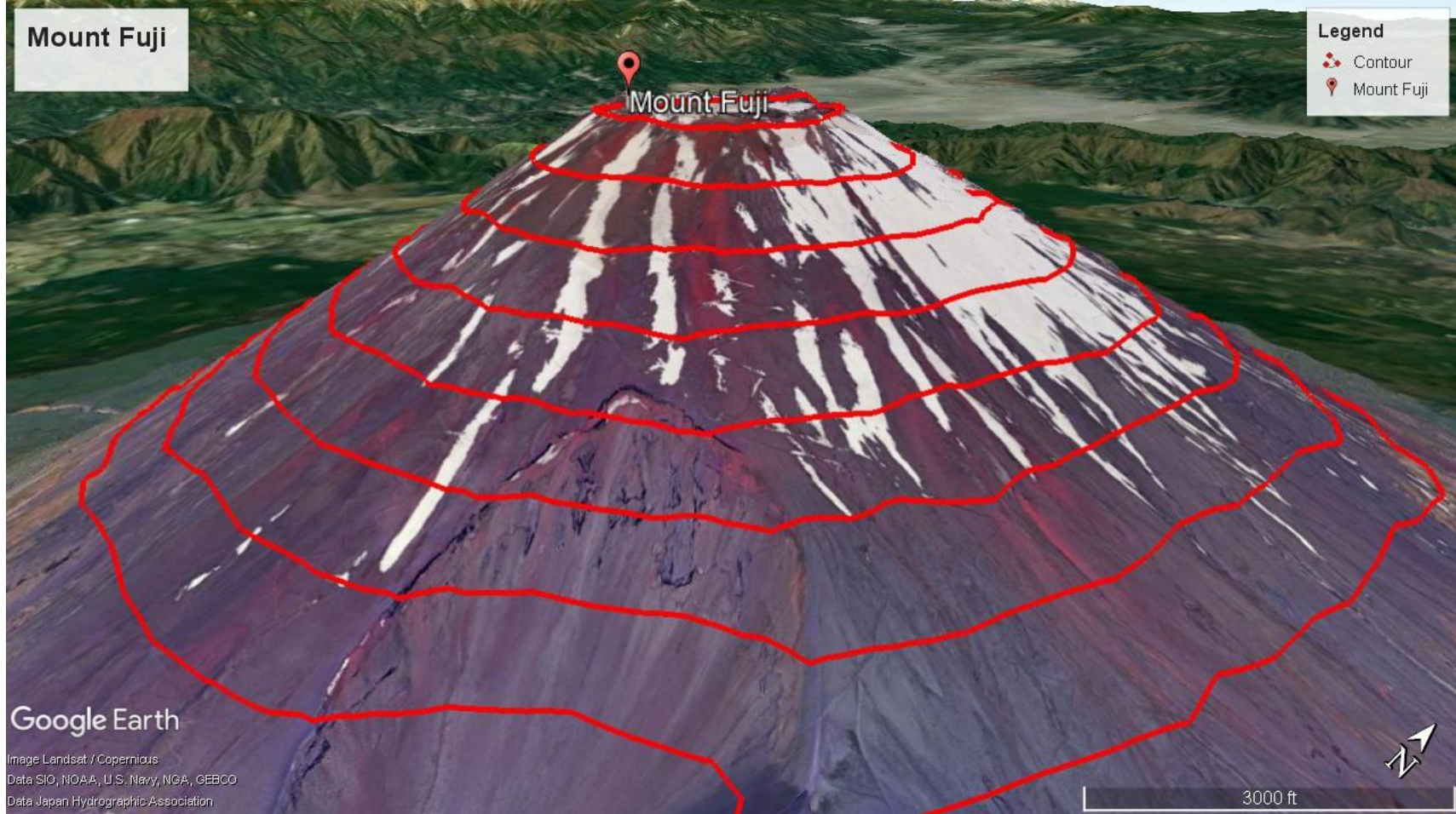


# 3D surface plot



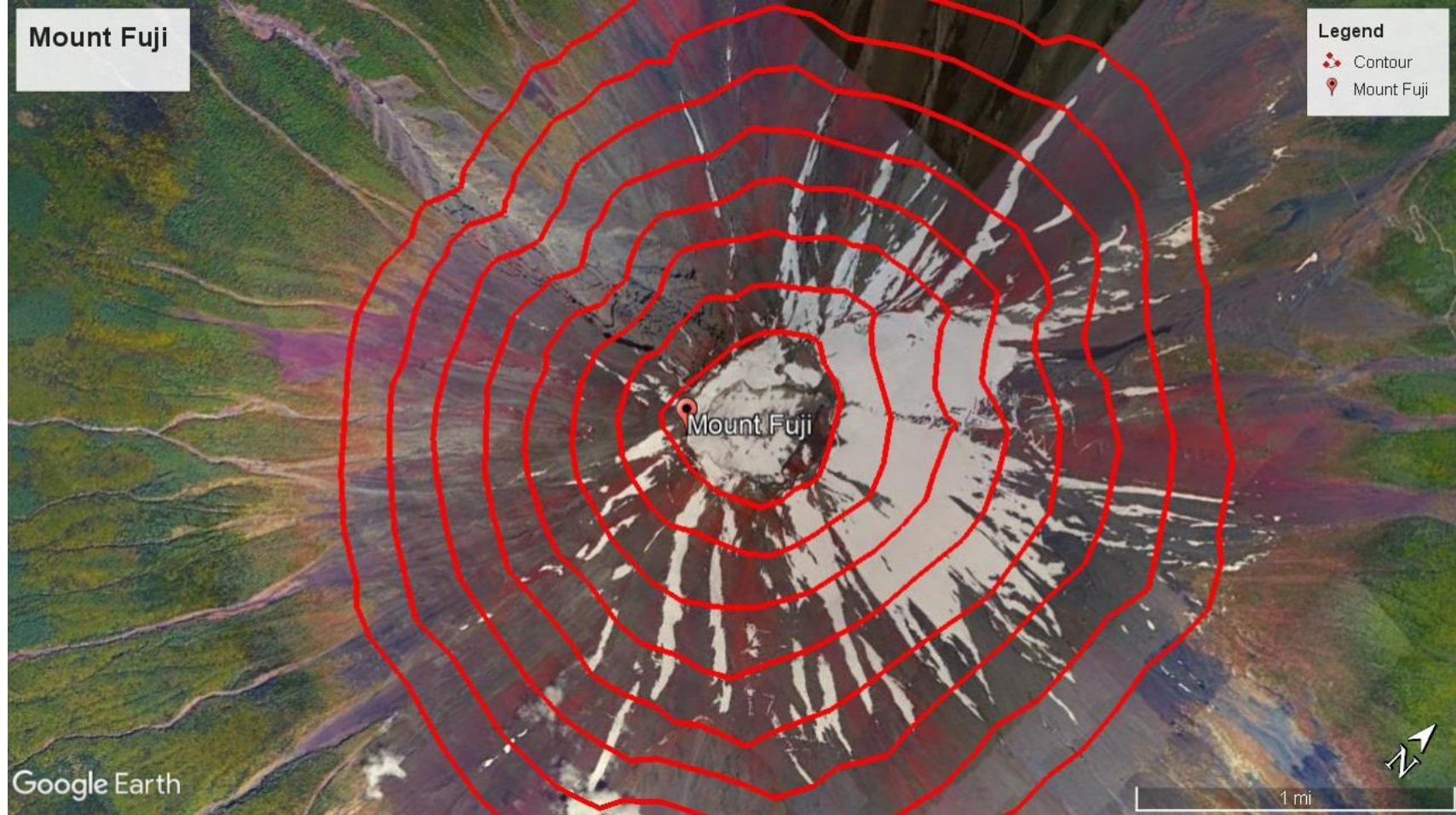
Alternative  
contour plot

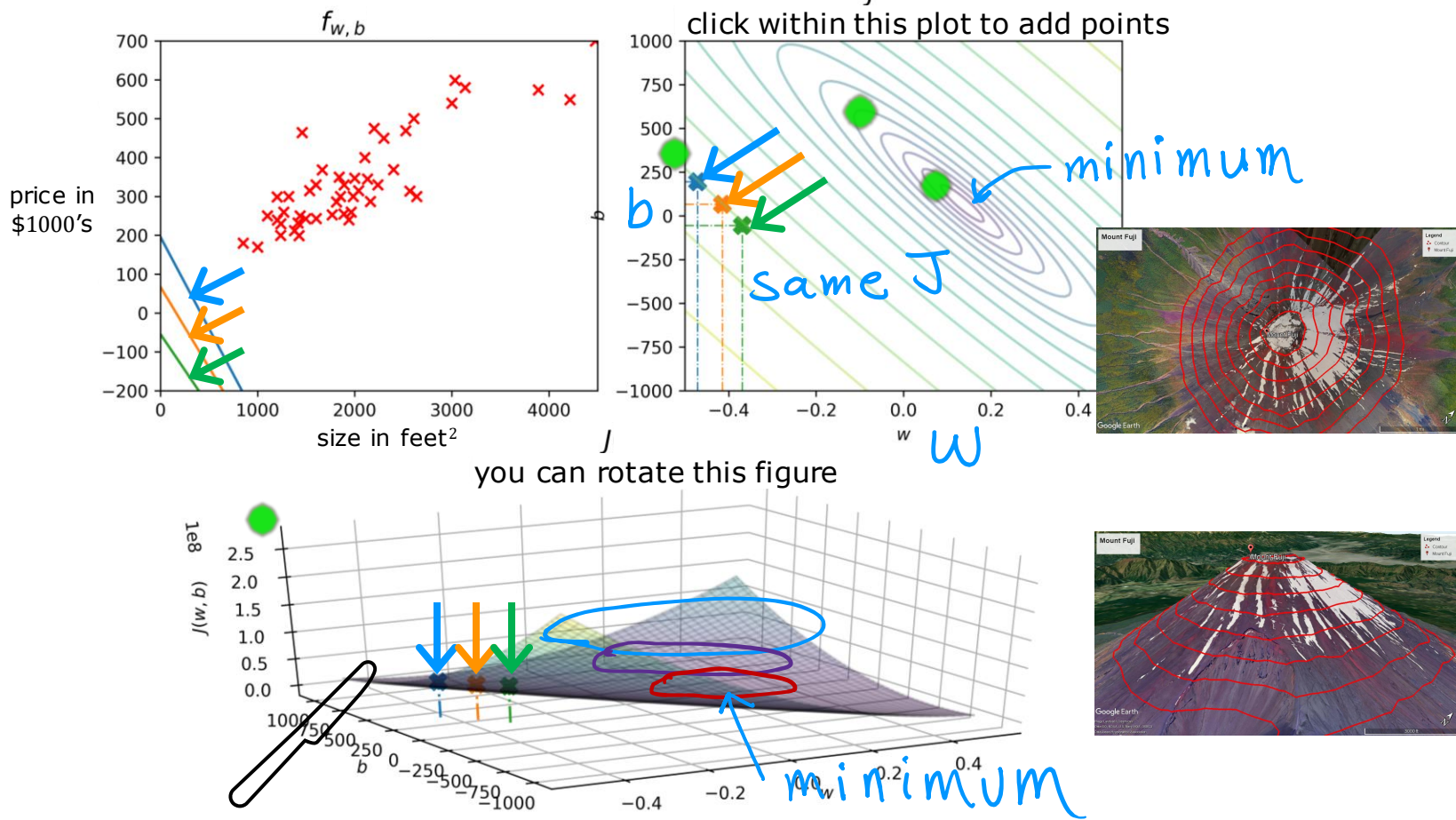
## Mount Fuji





## Mount Fuji

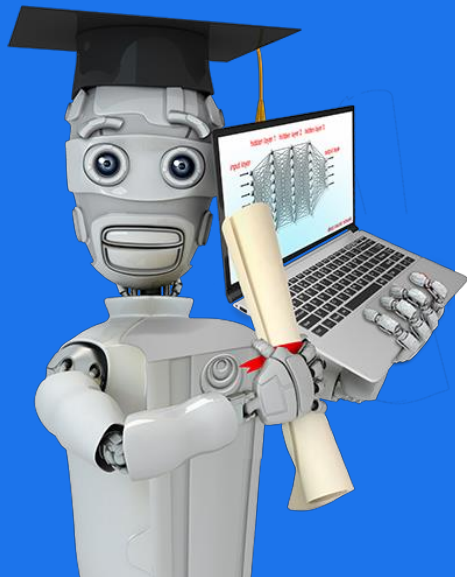






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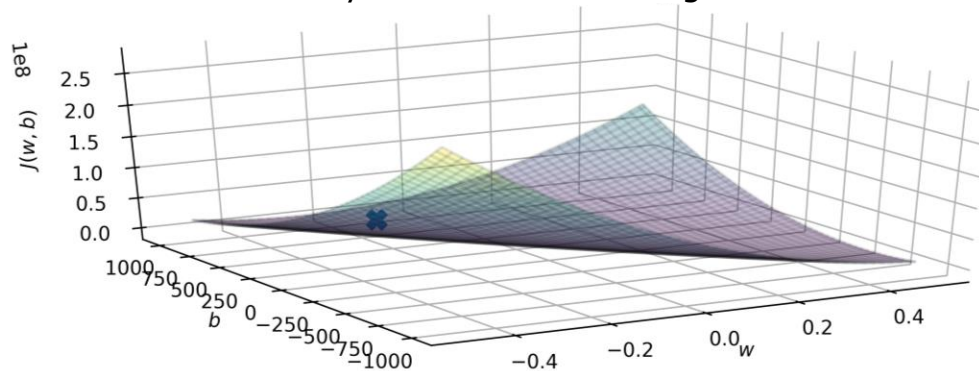
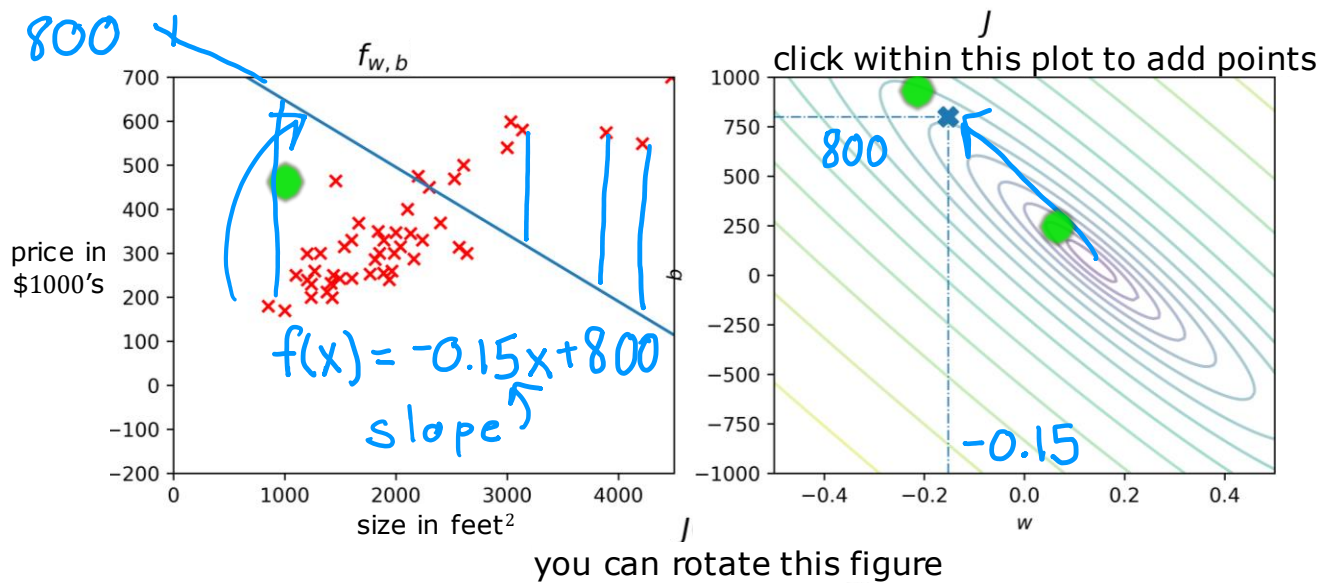
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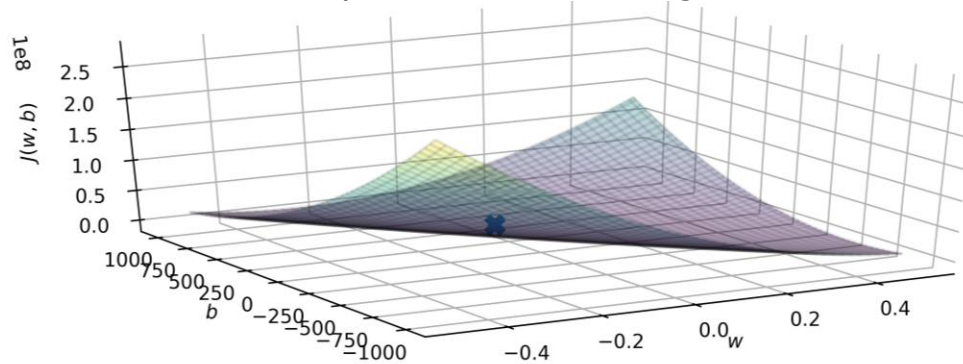
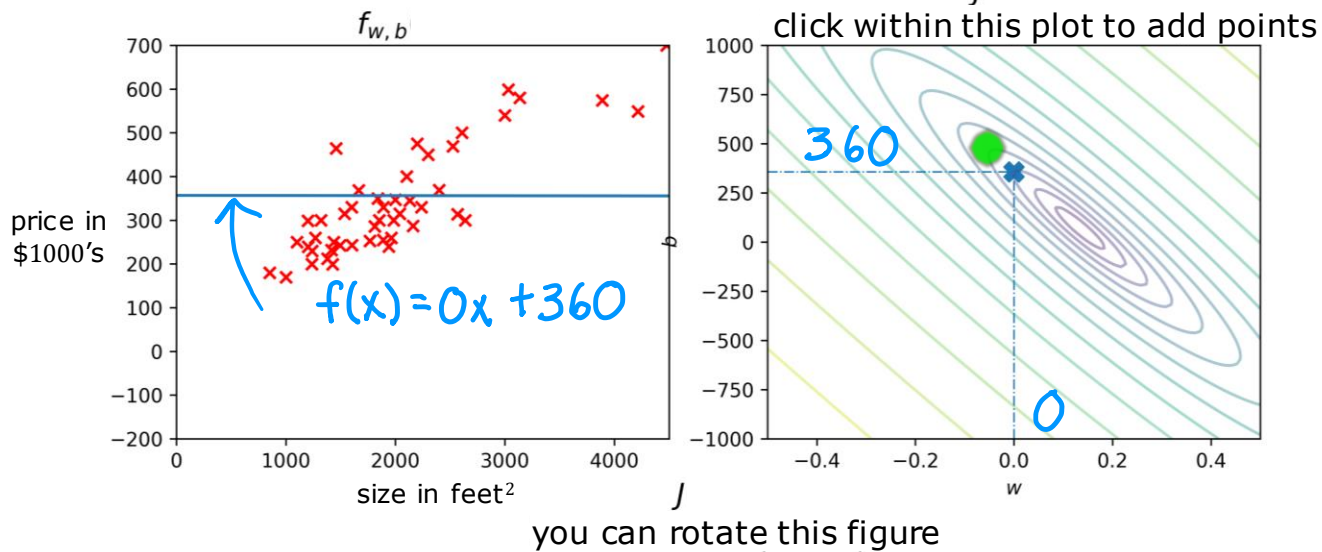


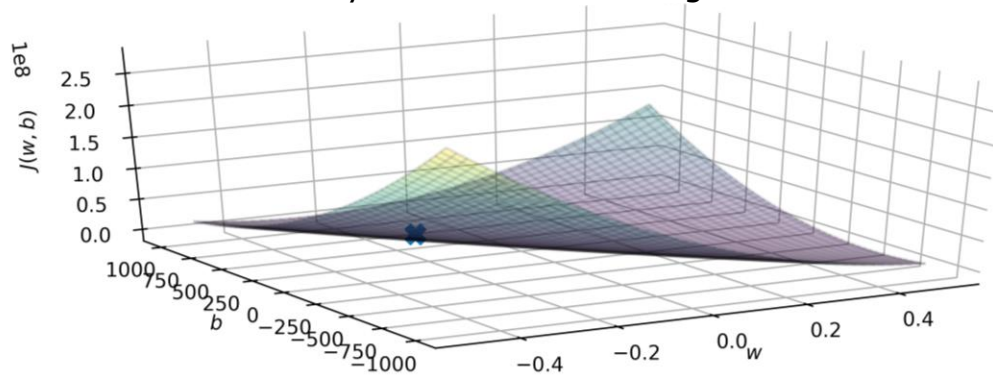
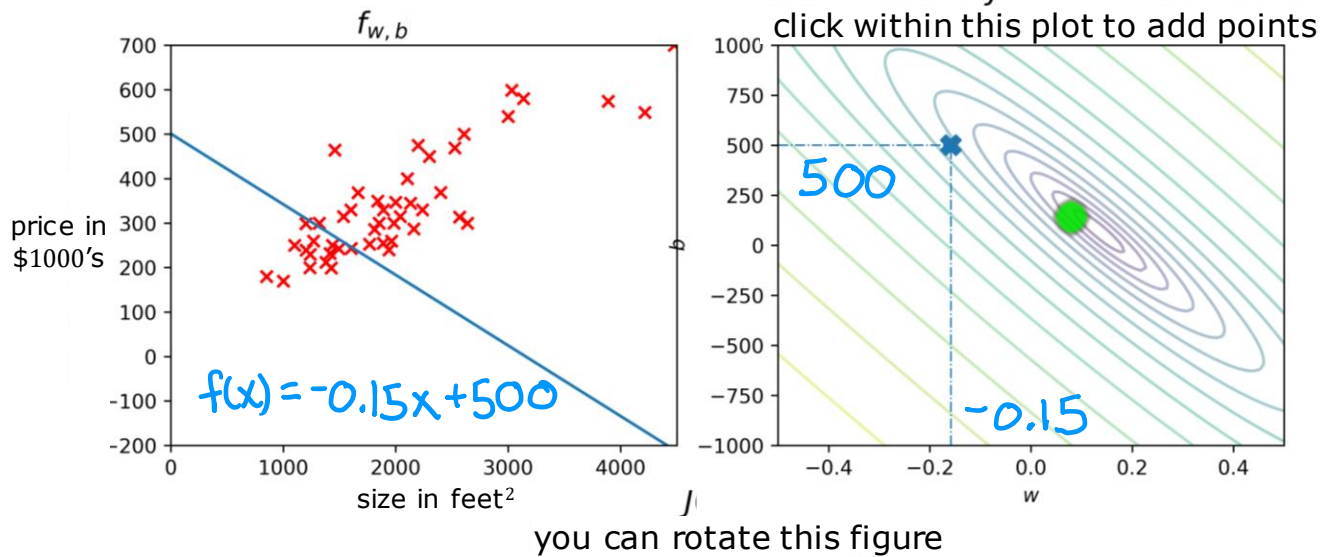
# Linear Regression with One Variable

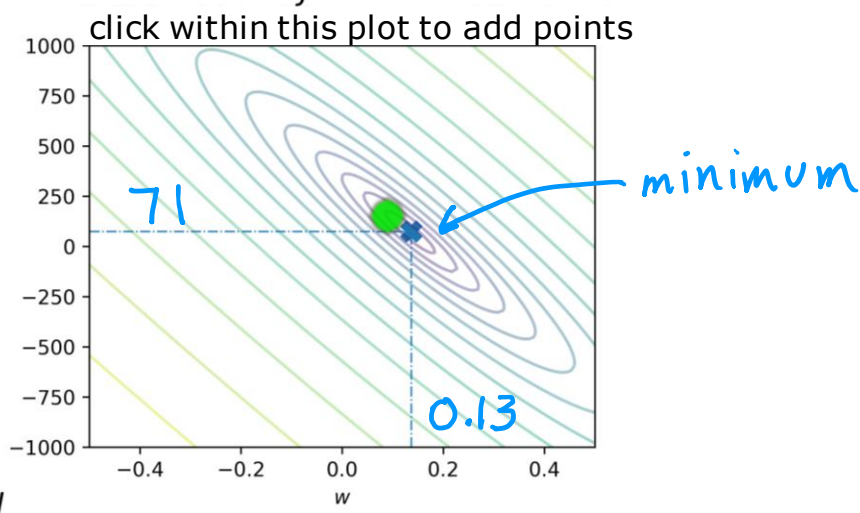
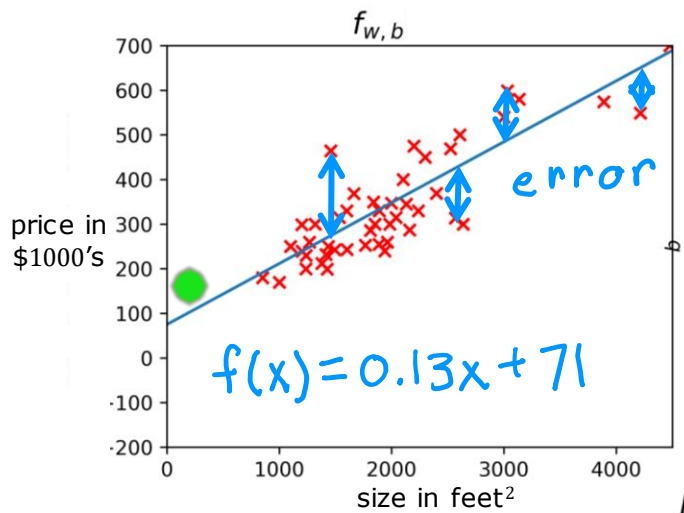
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## Visualization examples

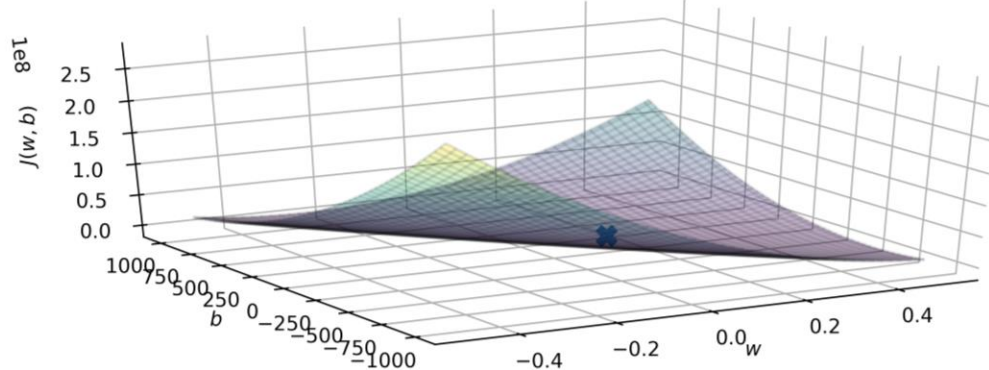






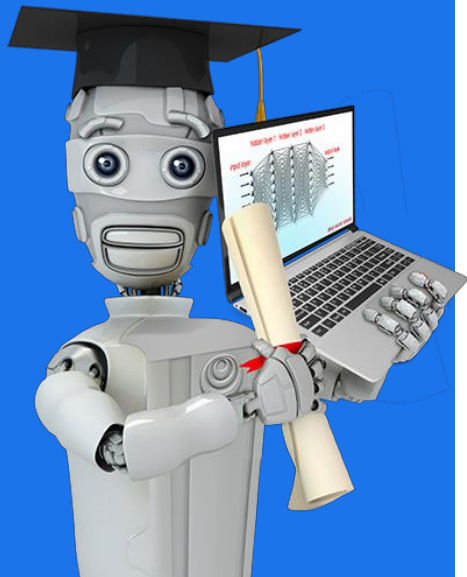


you can rotate this figure



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# Training Linear Regression

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## Gradient Descent

Have some function  $J(w, b)$  *for linear regression  
or any function*

Want  $\min_{w, b} J(w, b)$   $\min_{w_1, \dots, w_n, b} \underline{J(w_1, w_2, \dots, w_n, b)}$

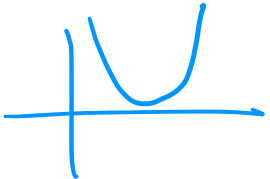
Outline:

Start with some  $w, b$  (set  $w=0, b=0$ )

Keep changing  $w, b$  to reduce  $J(w, b)$

Until we settle at or near a minimum

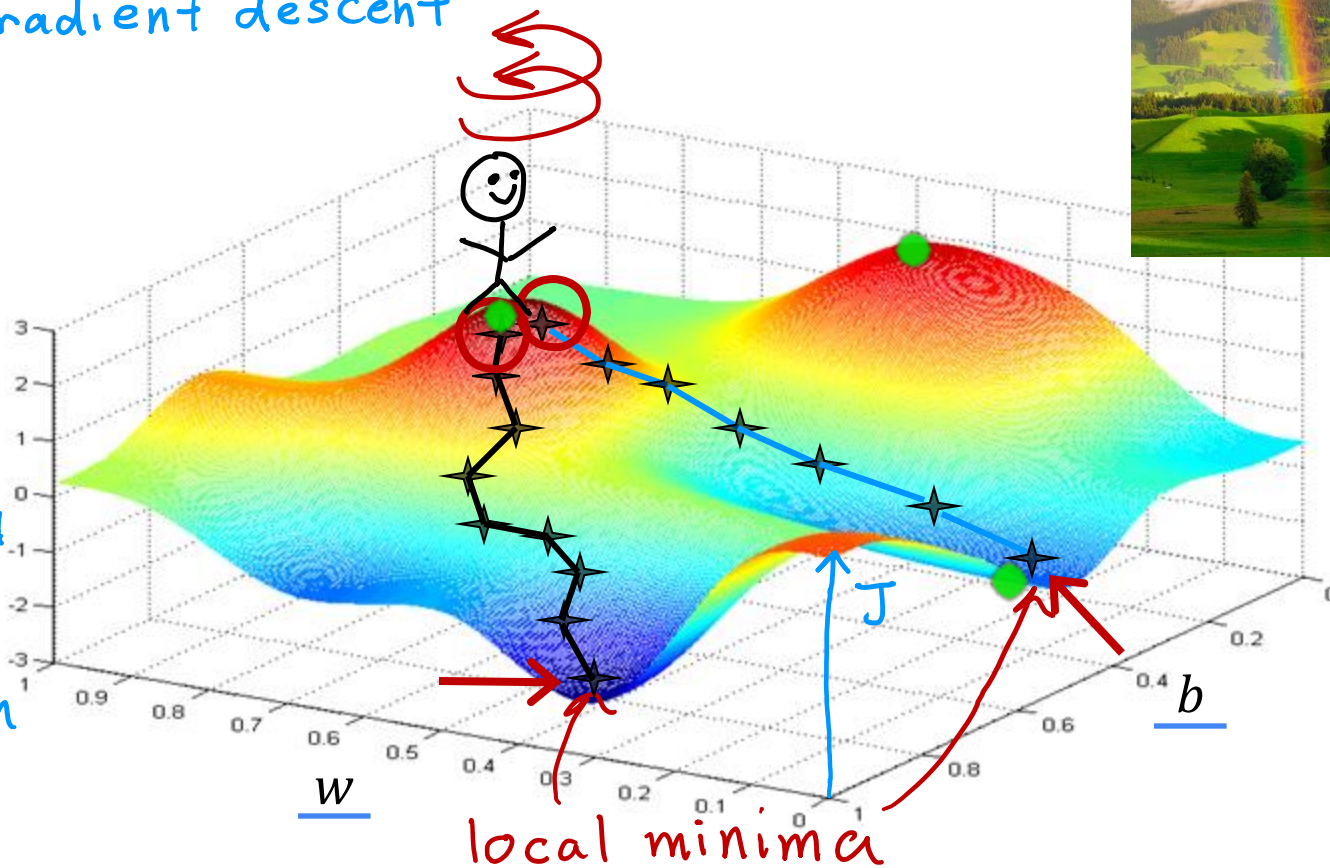
*may have  $>1$  minimum*

*J not always*  




gradient descent

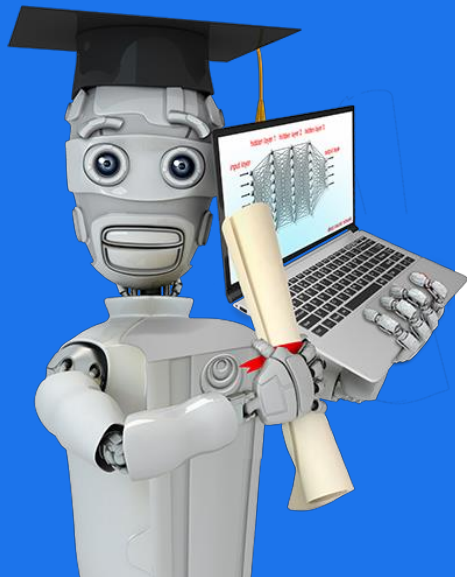
$J(w, b)$   
not squared  
error cost  
not linear  
regression





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# Training Linear Regression

---

## Implementing Gradient Descent

# Gradient descent algorithm

Repeat until convergence

$$\begin{cases} \underline{w} = w - \alpha \frac{\partial}{\partial w} J(w, b) \\ \underline{b} = b - \alpha \frac{\partial}{\partial b} J(w, b) \end{cases}$$

Learning rate  
Derivative

Simultaneously  
update  $w$  and  $b$

Assignment

$$a = c$$

$$a = a + 1$$

Code

Truth assertion

$$a = c$$

$$a = a + 1$$

Math

$$a == c$$

Correct: Simultaneous update

$$tmp\_w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$tmp\_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$w = tmp\_w$$

$$b = tmp\_b$$

Incorrect

$$tmp\_w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

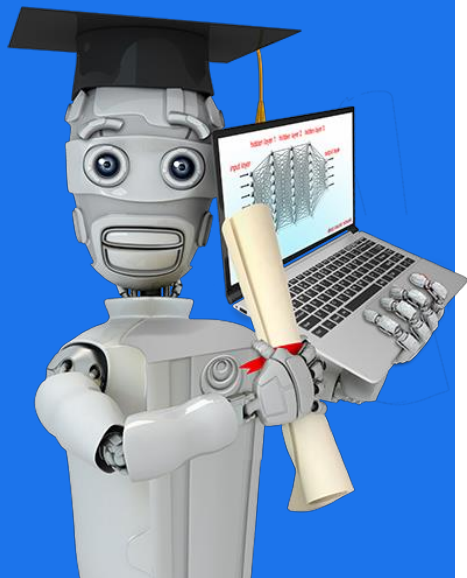
$$\underline{w} = tmp\_w$$

$$tmp\_b = b - \alpha \frac{\partial}{\partial b} J(\underline{w}, b)$$

$$\underline{b} = tmp\_b$$

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# Training Linear Regression

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## Gradient Descent Intuition

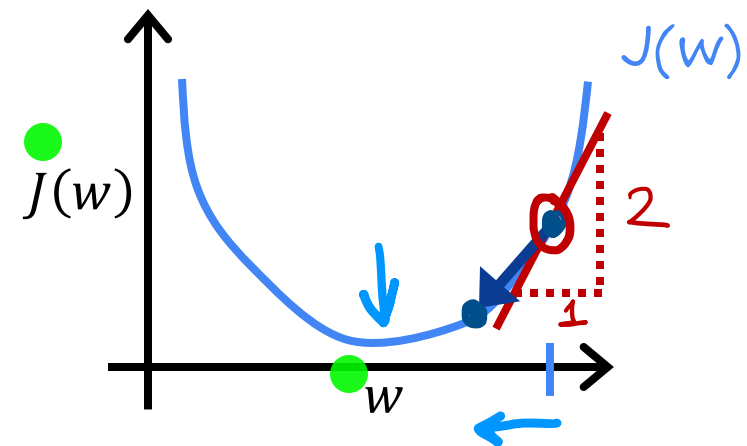
# Gradient descent algorithm

● repeat until convergence {

learning rate  $\alpha$  derivative

$$\begin{cases} \underline{w} = w - \alpha \frac{\partial}{\partial w} J(w, b) \\ \underline{b} = b - \alpha \frac{\partial}{\partial b} J(w, b) \end{cases}$$

$$J(w)$$
$$w = w - \alpha \frac{\partial}{\partial w} J(w)$$
$$\underline{\min_w} J(w)$$



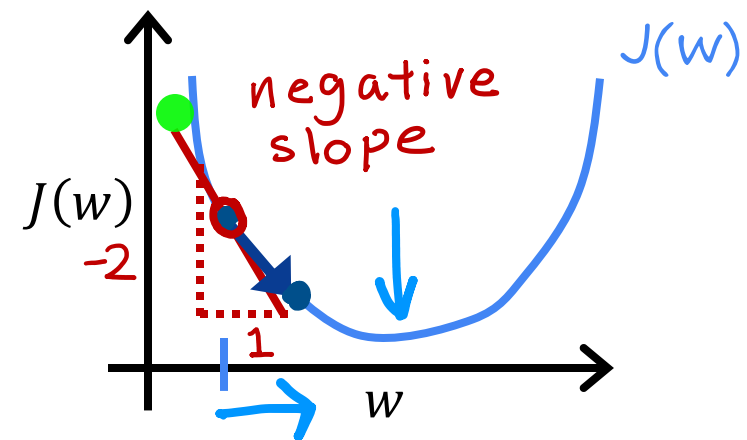
$$w = w - \alpha \underbrace{\frac{d}{dw} J(w)}_{> 0}$$

$$w = w - \underline{\alpha} \cdot (\text{positive number})$$

$$\underline{\frac{d}{dw} J(w)} < 0$$

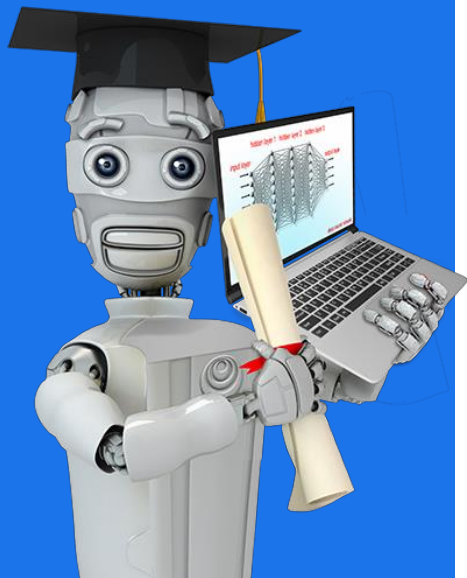
$$w = \underline{w} - \alpha \cdot (\text{negative number})$$

↑                      ↑



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# Training Linear Regression

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## Learning Rate

$$w = w - \alpha \frac{d}{dw} J(w)$$

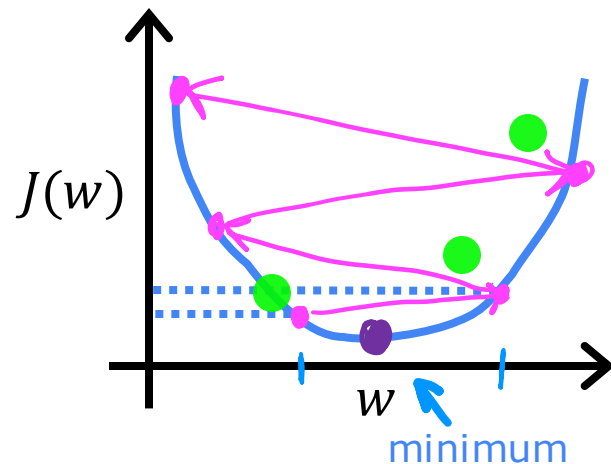
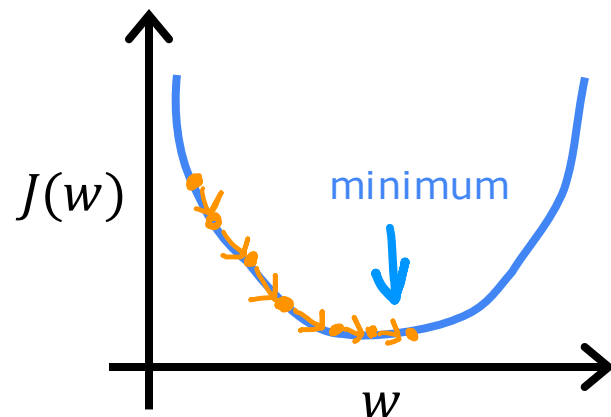
If  $\alpha$  is too small...

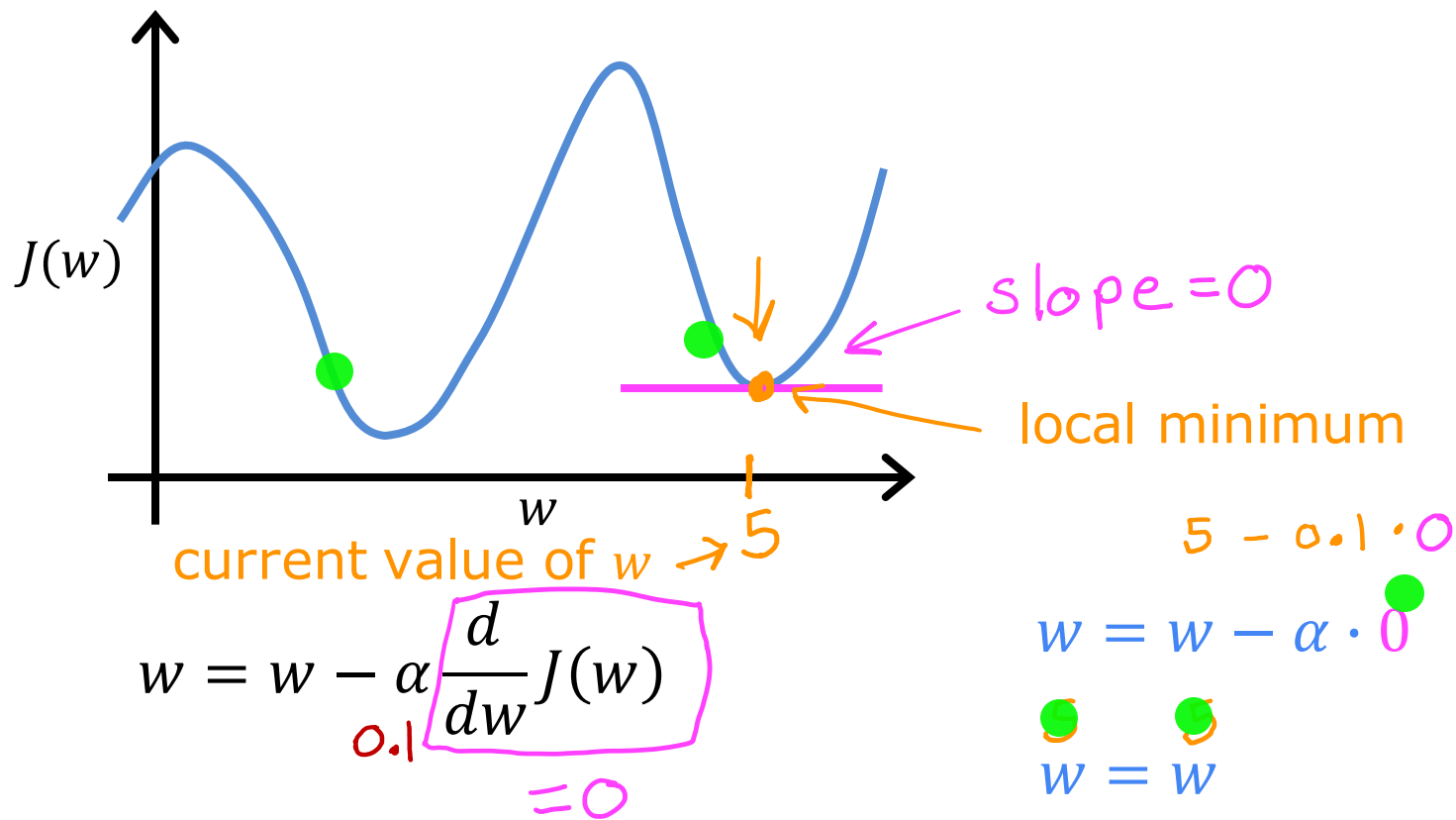
Gradient descent may be slow.

If  $\alpha$  is too large...

Gradient descent may:

- Overshoot, never reach minimum
- Fail to converge, diverge







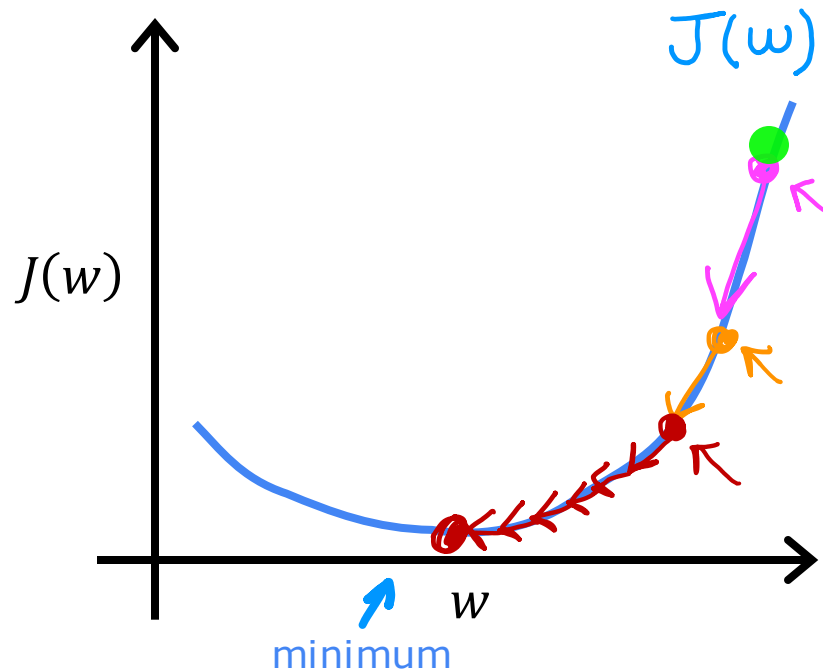
Can reach local minimum with fixed learning rate  $\alpha$

$$w = w - \underbrace{\alpha}_{\text{smaller}} \underbrace{\frac{d}{dw} J(w)}_{\text{not as large}} \underbrace{J(w)}_{\text{large}}$$

Near a local minimum,

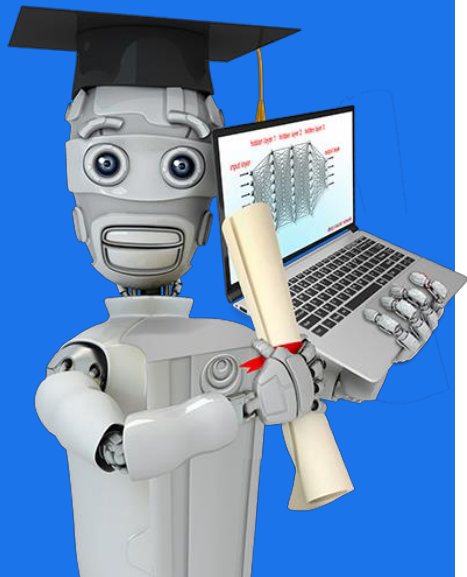
- Derivative becomes smaller
- Update steps become smaller

Can reach minimum without decreasing learning rate  $\alpha$



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# Training Linear Regression

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## Gradient Descent for Linear Regression

## Linear regression model

$$f_{w,b}(x) = wx + b$$

## Cost function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

## Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})x^{(i)}$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

}

next slide  
is optional!

(Optional)

$$\frac{\partial}{\partial w} J(w, b) = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2 = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) \cancel{2} x^{(i)} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$\frac{\partial}{\partial b} J(w, b) = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2 = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) \cancel{2} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

no  $x^{(i)}$

# Gradient descent algorithm

$$\frac{\partial}{\partial w} J(w, b)$$

repeat until convergence {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

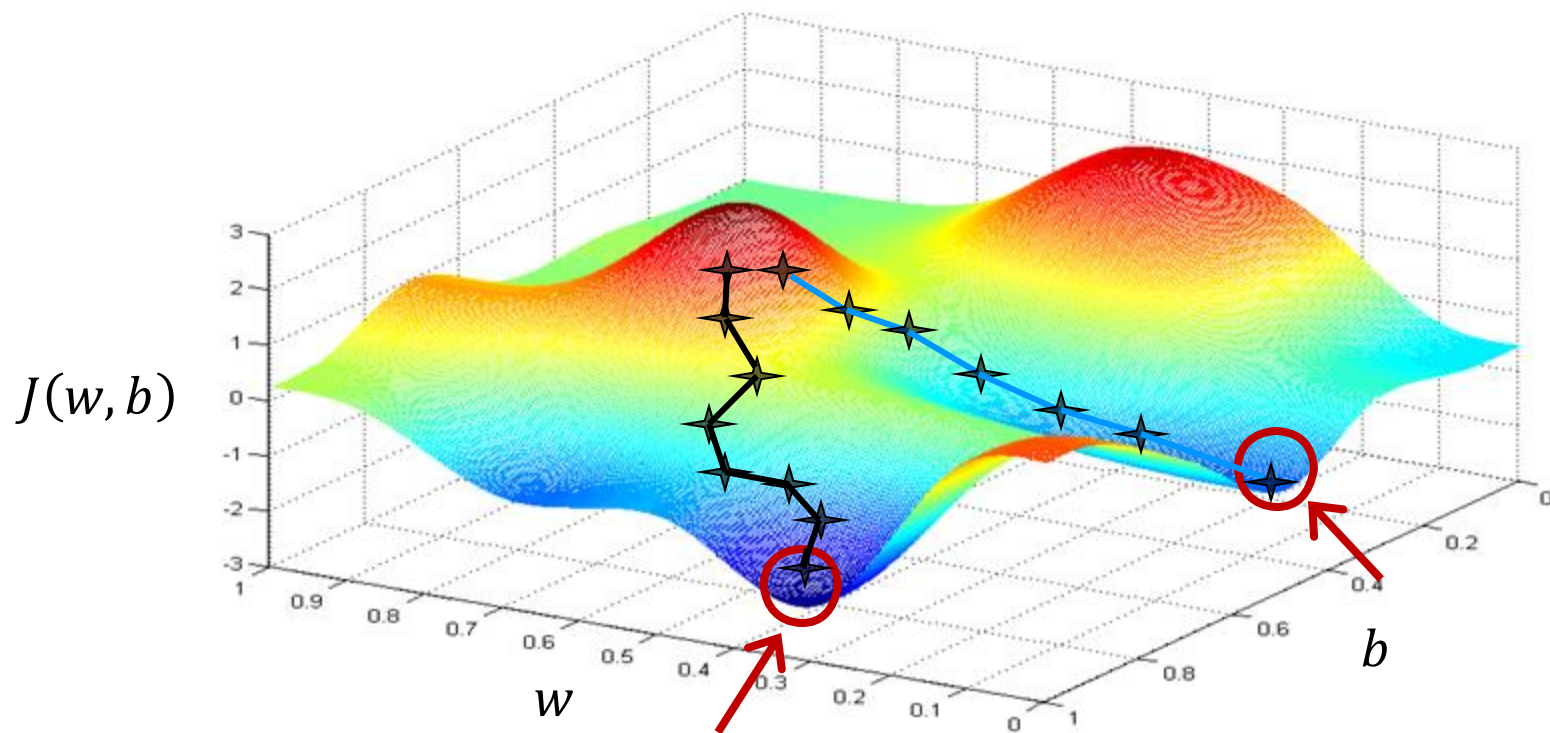
}

$$\frac{\partial}{\partial b} J(w, b)$$

Update  
w and b  
simultaneously

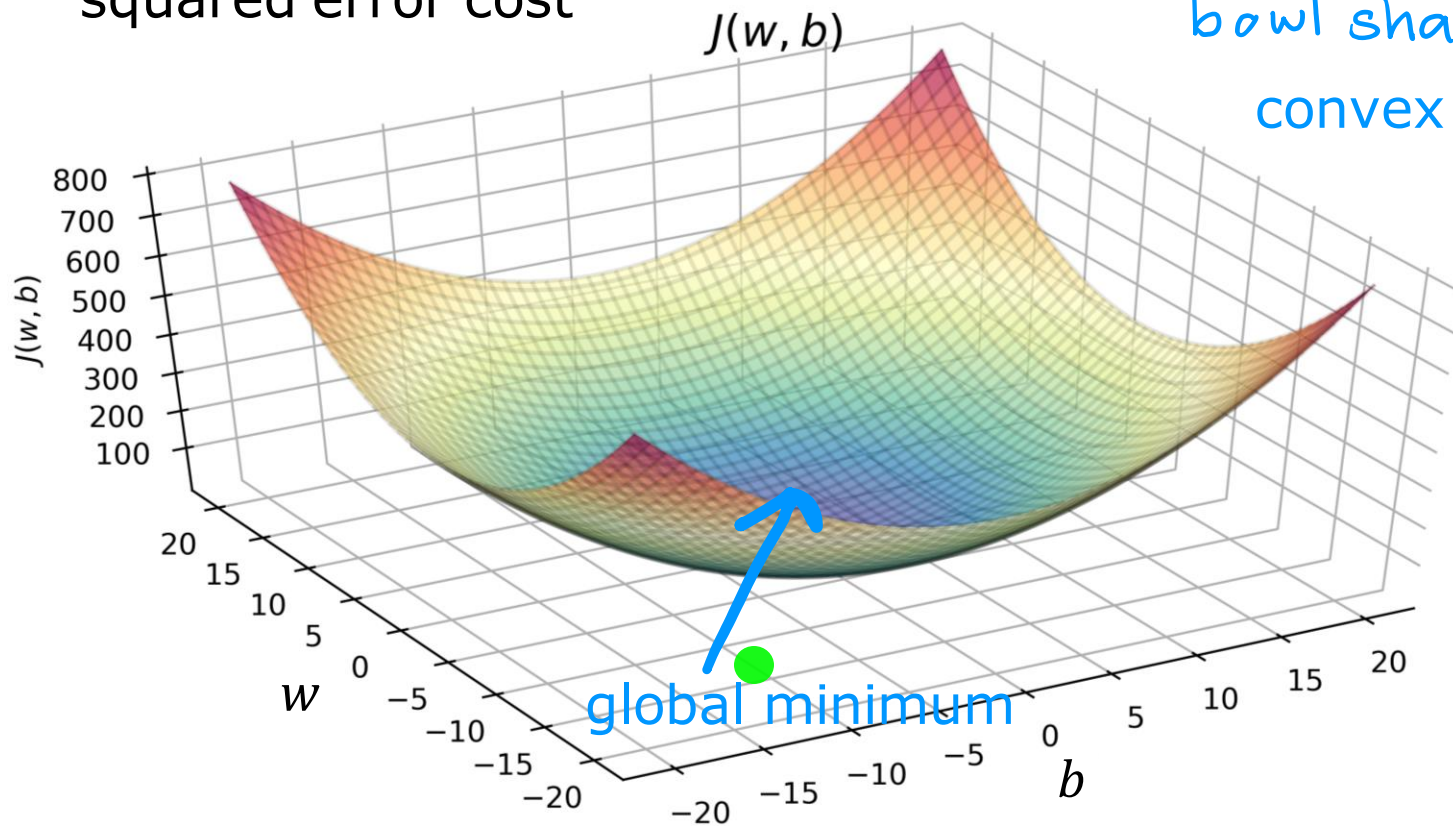
$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

## More than one local minimum



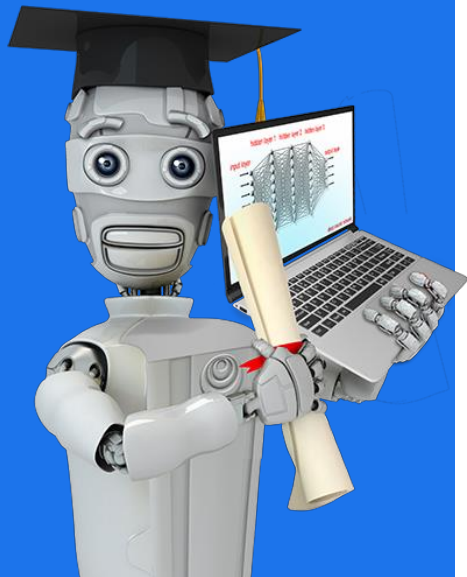
squared error cost

bowl shape   
convex function



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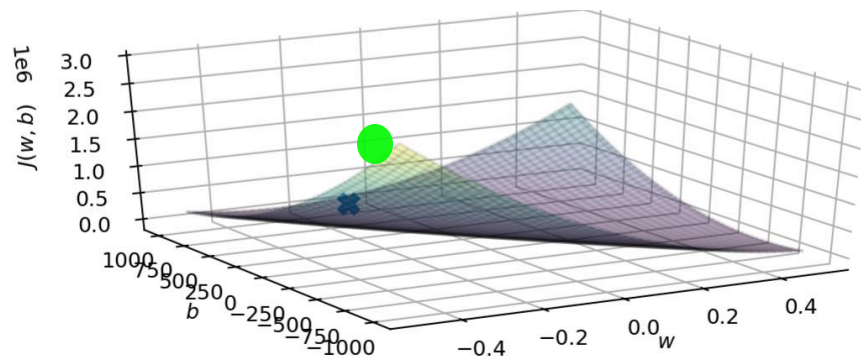
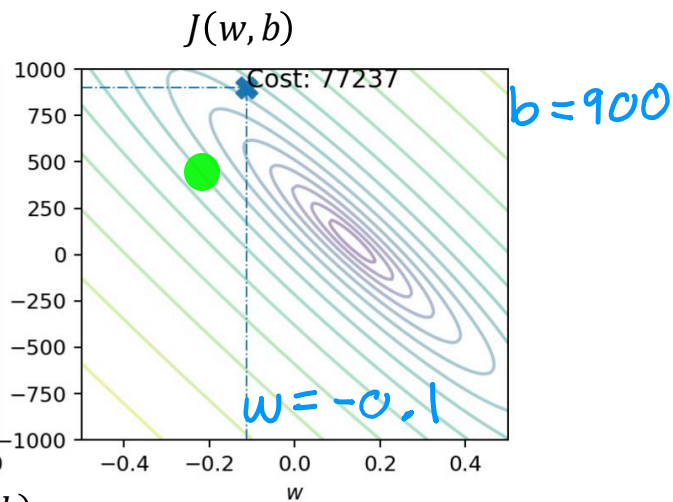
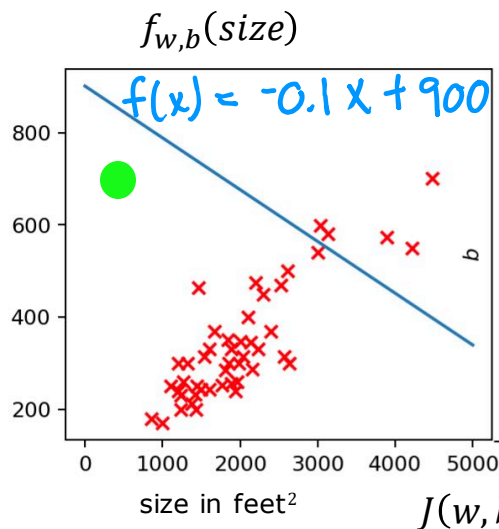
# Training Linear Regression

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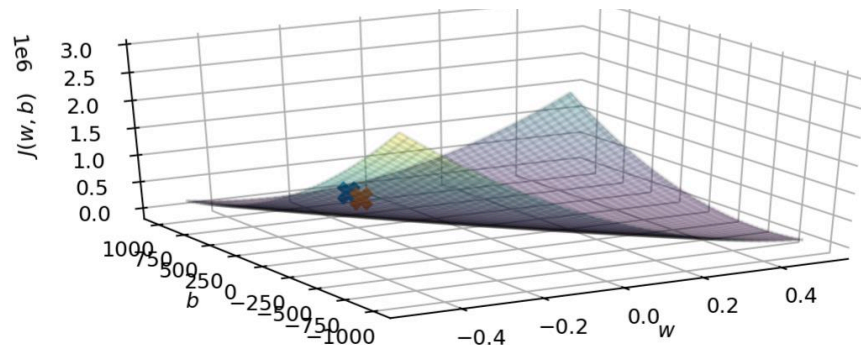
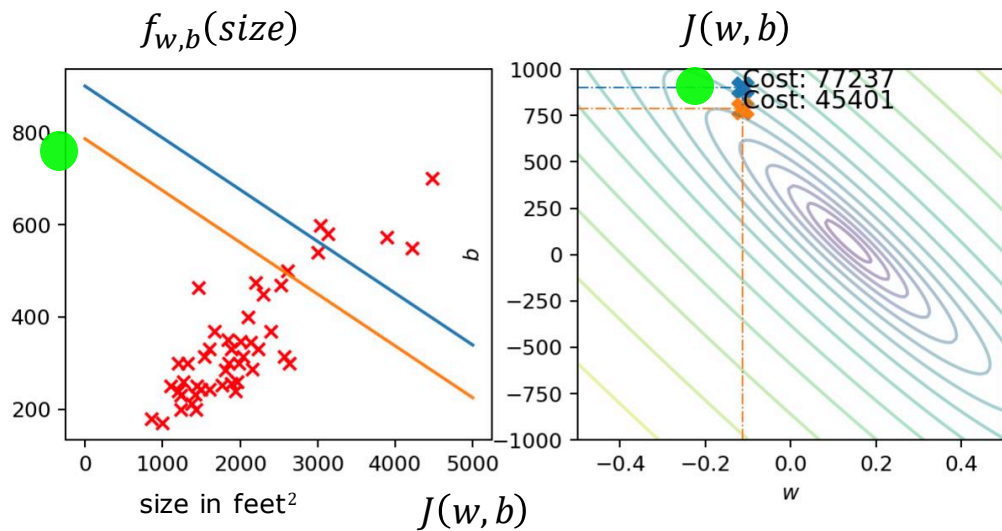
## Running Gradient Descent

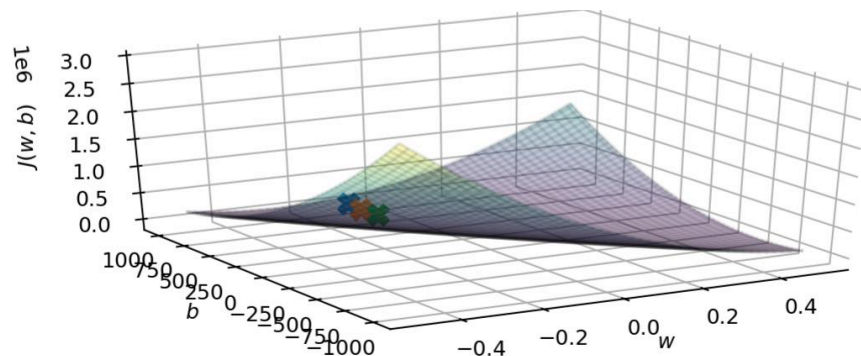
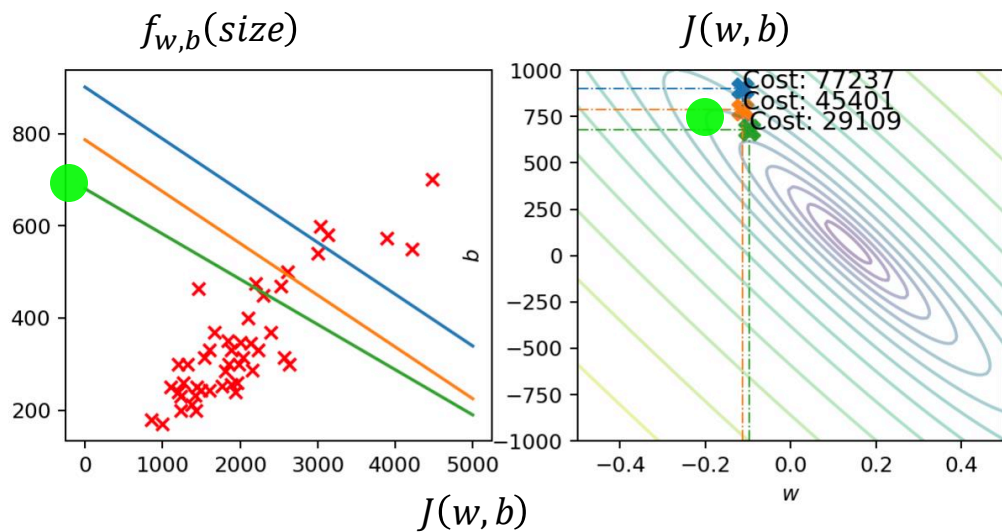


price in  
\$1000's

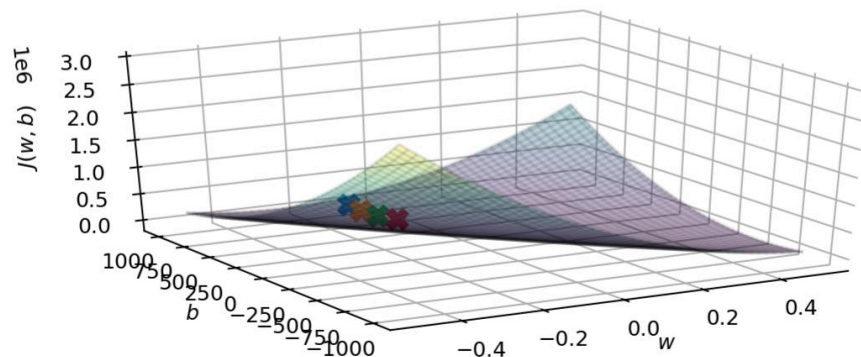
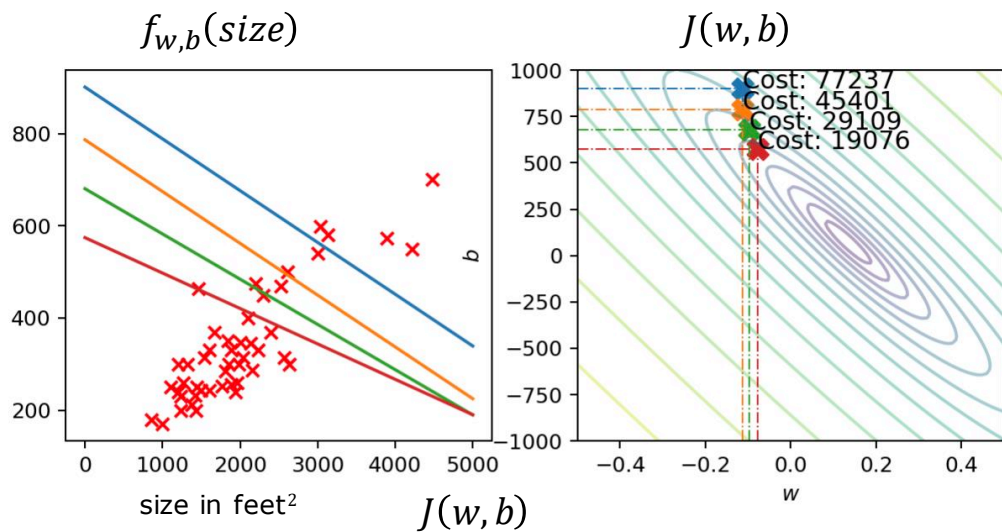


price in  
\$1000's

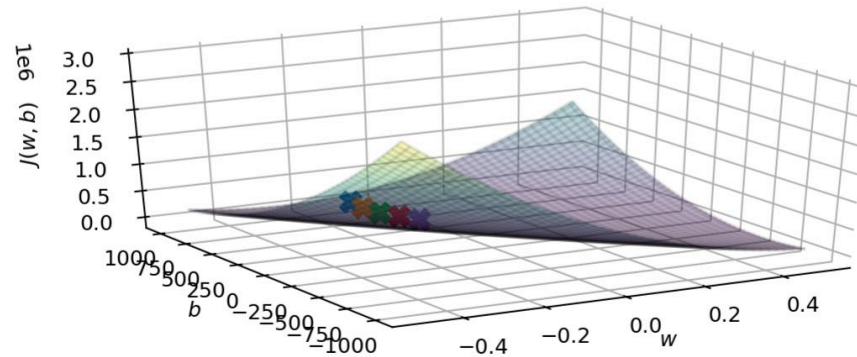
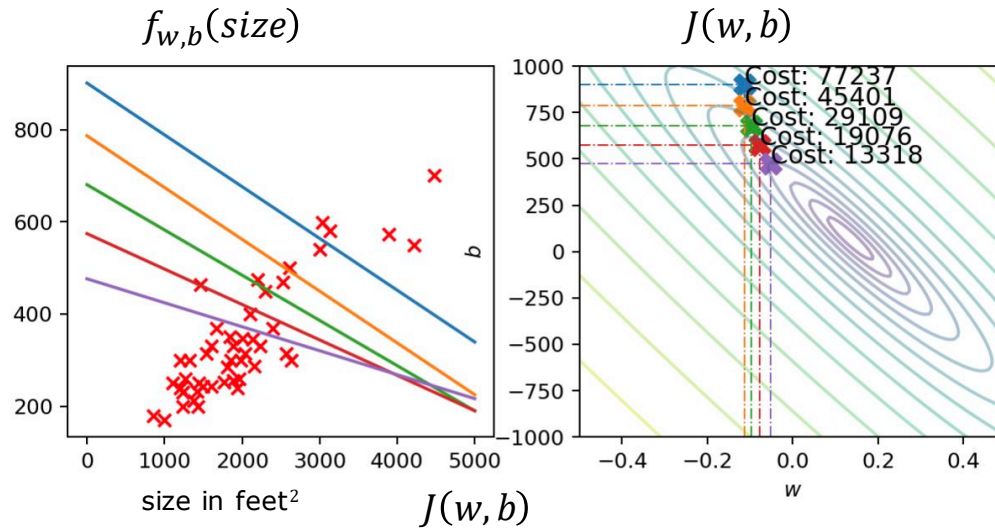




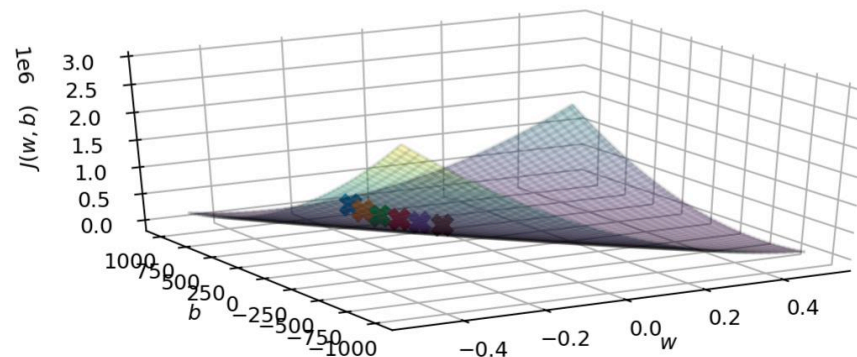
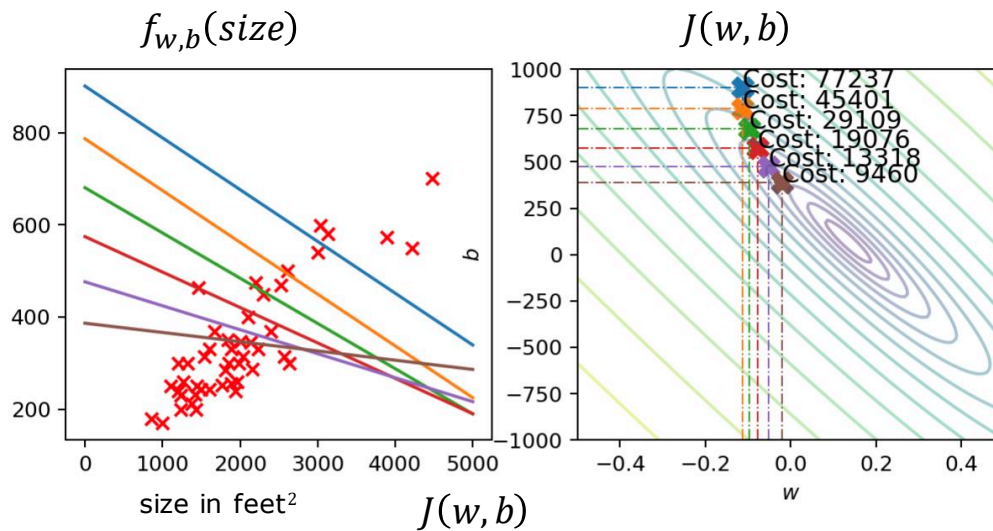
price in  
\$1000's



price in  
\$1000's

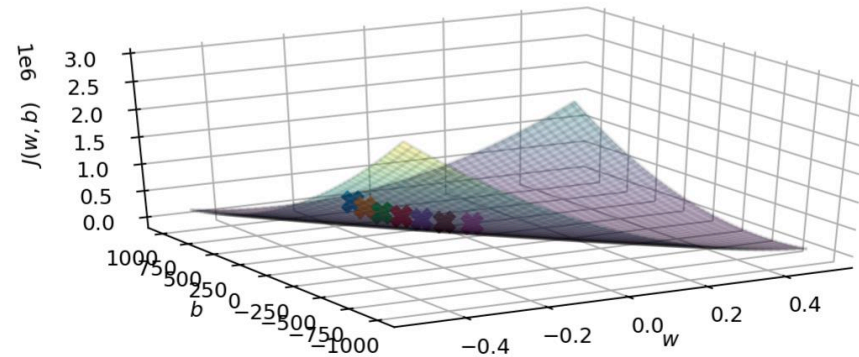
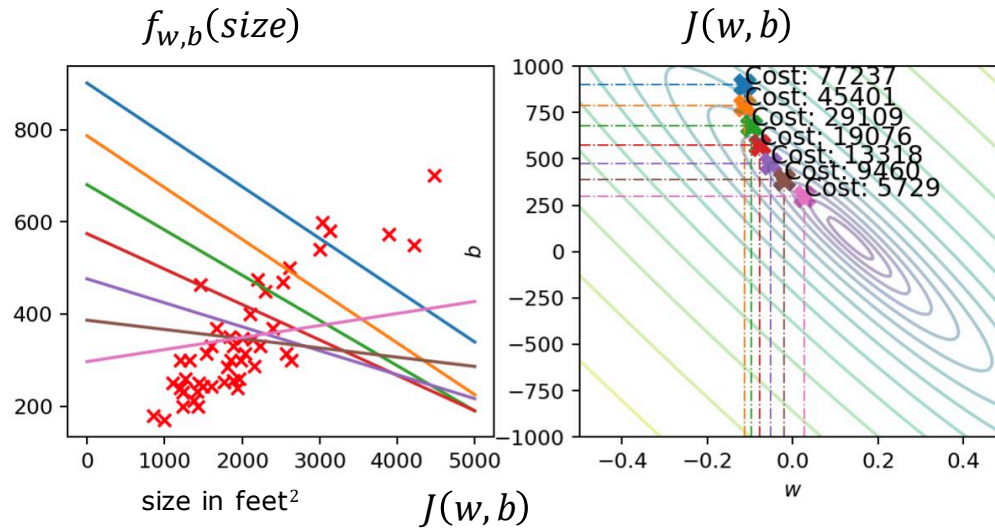


price in  
\$1000's

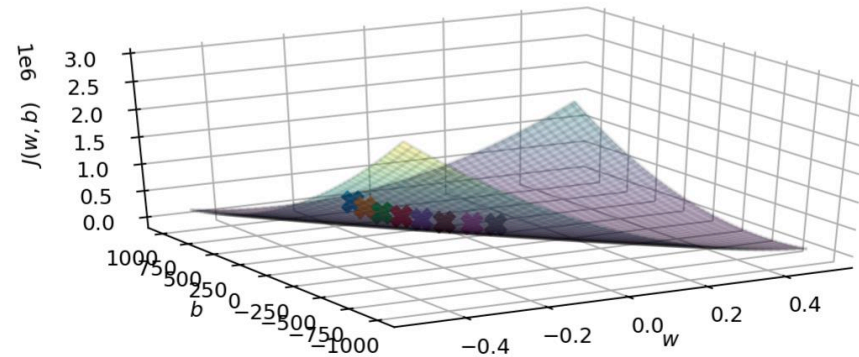
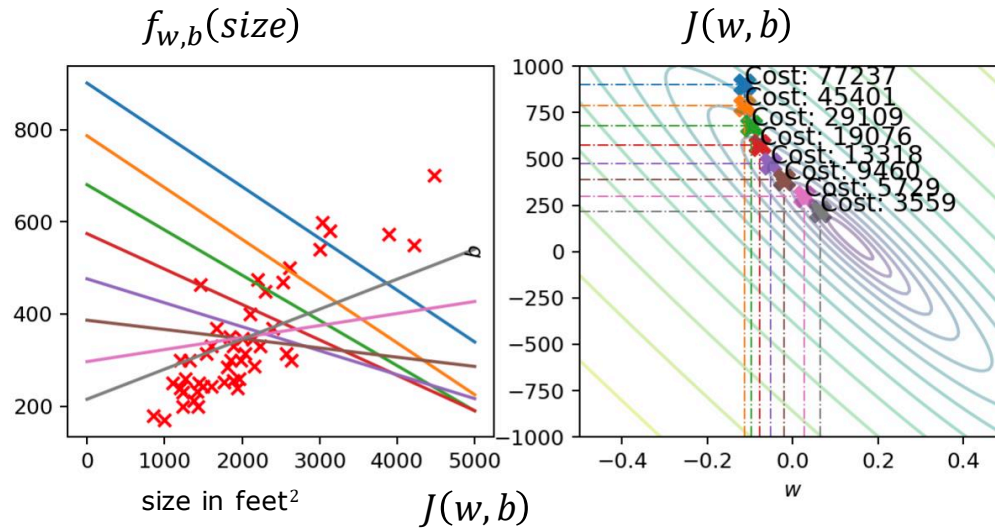




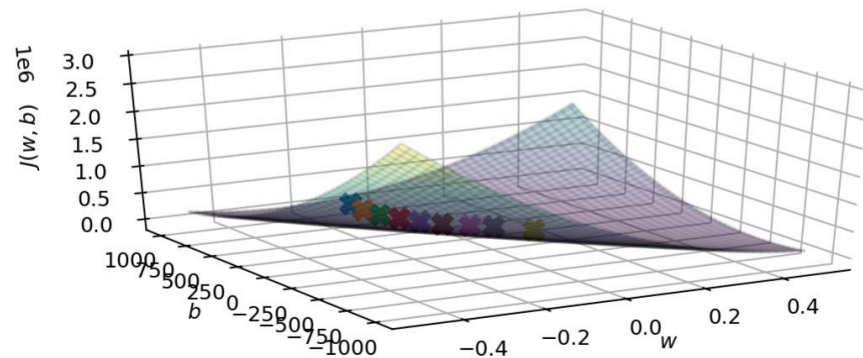
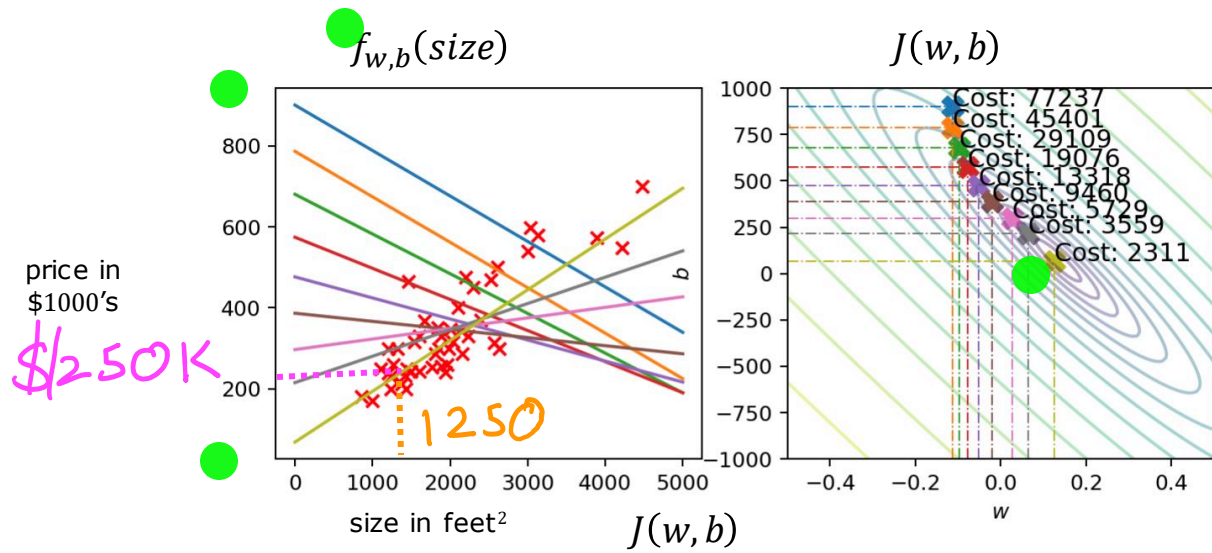
price in  
\$1000's



price in  
\$1000's







# "Batch" gradient descent

"Batch": Each step of gradient descent uses all the training examples.

other gradient descent: subsets

|      | $x$<br>size in feet <sup>2</sup> | $y$<br>price in \$1000's |
|------|----------------------------------|--------------------------|
| (1)  | 2104                             | 400                      |
| (2)  | 1416                             | 232                      |
| (3)  | 1534                             | 315                      |
| (4)  | 852                              | 178                      |
| ...  | ...                              | ...                      |
| (47) | 3210                             | 870                      |

$m = 47$

$$\sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

